

MASTER'S DEGREE IN COMPUTER SCIENCE

# STATISTICS FOR MANAGEMENT

Course Notes / 2026–2027

OBSERVATIONS / STRUCTURE / INFERENCE



DATA ANALYSIS · STATISTICAL COMPUTING · CLUSTERING

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# Introduction

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Statistics turns observations into evidence for decisions. In management applications, it supports the description of customers and organizations, comparisons between groups, the study of relationships between variables, and the discovery of structure in complex datasets. Sound analysis connects a clearly stated question with suitable data, mathematical methods and an interpretation that respects the limits of the evidence.

Descriptive statistics summarizes the observations available through tables, graphs and numerical measures. Statistical inference extends conclusions from a sample to a wider population while accounting for uncertainty. The descriptive methods developed here provide the foundation for that distinction: a pattern in observed data must first be understood before its broader implications can be assessed.

These notes develop measurement scales, frequency distributions, measures of location and variability, bivariate association and simple linear regression. Linear algebra then provides the language for multivariate data, covariance matrices and distance measures. Statistical computing in R connects the formulas with practical analysis, and cluster analysis brings these ideas together to identify groups of similar observations.

## Using the notes

Each chapter introduces its central concepts, explains the notation used in significant formulas, and ends with a compact review. Worked examples connect calculation with interpretation; schematic diagrams illustrate principles without representing empirical datasets. The final reference chapter collects the principal formulas, and the mathematical toolkit supplies supporting algebra and calculus.

Population summaries use  $N$ ,  $\mu$ ,  $\sigma^2$  and  $\rho$  where appropriate; sample estimates use  $n$ ,  $\bar{x}$ ,  $s^2$  and  $r$ . Matrix and clustering sections state their dimensions locally. In particular, the divisor  $N$  in a descriptive population variance differs from the divisor  $n - 1$  in the usual sample variance returned by R.

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# Statistical Foundations and Data Structures

Statistics begins with a question and a precise description of the units and variables being observed. Measurement scales determine meaningful comparisons, while data matrices and frequency distributions organize the evidence for analysis.

## 1.1 Statistics as a Problem-Solving Process

Statistics is a problem-solving discipline whose formulas serve a defined analytical question. A useful organizing framework is the PPDAC cycle: **P**roblem, **P**lan, **D**ata, **A**nalysis, and **C**onclusion. The cycle emphasizes that statistical analysis starts before computation: the question determines the plan, the plan determines what data are required, and the analysis must ultimately return to the original problem.

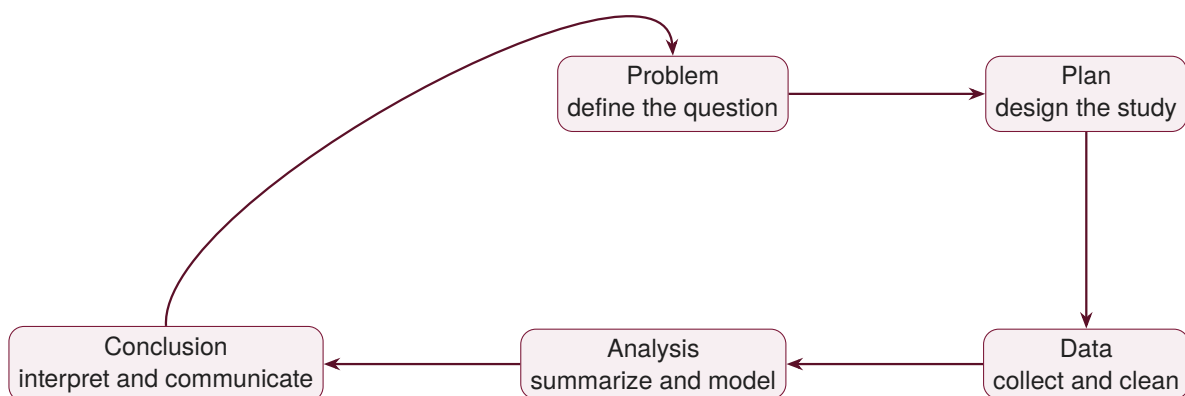


Figure 1.1: The PPDAC cycle connects the statistical question, study design, data collection, analysis and interpretation.

### 1.1.1 Descriptive and inferential statistics

#### Definition: Descriptive / exploratory statistics

Descriptive statistics summarizes observed data using numerical measures, tables, plots and structural features. The summaries refer directly to the observed units, whether they form a population or a sample.

**Definition: Statistical inference**

The observed data are treated as a sample from a larger population. Statistical methods are then used to reason beyond the sample while explicitly accounting for uncertainty, typically through probability.

Descriptive statistics and probability provide the foundations for inference: inference requires an understanding of what the data say directly and of how uncertainty is represented.

**1.1.2 Univariate and multivariate analysis**

A **univariate** analysis studies one characteristic at a time. For example, family wealth may be studied only through income. A **multivariate** analysis studies several characteristics and the relationships among them—for example employment type, educational qualification, assets, income and property in a household survey. Joint analysis of several variables underlies covariance matrices and clustering.

**Takeaway**

A useful analytical workflow is: learn the mathematical relationship, understand its meaning, use R to compute it, and interpret the software output. Programming is presented as a tool for understanding statistical concepts rather than as a separate objective.

**1.2 Statistical Units, Variables and the Data Matrix**

A **statistical unit** is the object on which one or more variables are observed: a person, a car, a shop, a company, or another unit relevant to the study. Let  $n$  denote the number of units. A variable is conventionally written using a capital letter such as  $X$ , and the value observed for unit  $i$  is written  $x_i$ .

If a variable can assume  $k$  distinct categories or levels, the set of possible manifestations can be represented as

$$\mathcal{X} = \{c_1, c_2, \dots, c_k\}. \quad (1.1)$$

- $\mathcal{X}$  set of possible categories of the variable  $X$ ;
- $c_j$  the  $j$ th category, distinct from an observation  $x_i$ ;
- $k$  number of distinct categories or levels.

A **data matrix** places statistical units in rows and variables in columns. The UK Labour Force Survey example illustrates variables such as age, benefits, hours worked per week, months continuously employed, gross weekly pay and arrival year. The same example also emphasizes that real data matrices may contain missing or not-applicable values.

Table 1.1: Conceptual structure of a data matrix.

|          | $X_1$    | $X_2$    | $\dots$  | $X_p$    |
|----------|----------|----------|----------|----------|
| Unit 1   | $x_{11}$ | $x_{12}$ | $\dots$  | $x_{1p}$ |
| Unit 2   | $x_{21}$ | $x_{22}$ | $\dots$  | $x_{2p}$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\ddots$ | $\vdots$ |
| Unit $n$ | $x_{n1}$ | $x_{n2}$ | $\dots$  | $x_{np}$ |

For the Labour Force Survey example, the statistical unit is a UK resident, the reference population is UK residents, and the phenomenon under study is employment. Variables may describe sex, age, years of

work, marital status, nationality, family size, working conditions, weekly working hours, commuting, illness and many other characteristics.

A means-of-transport variable illustrates a category set: car/van/minibus or works van; motorbike/moped/scooter; bicycle; bus/coach/private bus; rail; underground; walking; and other methods of travelling. The example reinforces that categories are the admissible manifestations of a variable.

## 1.3 Measurement Scales and Variable Types

Correct statistical analysis depends on measuring each phenomenon with an appropriate scale. Four classical measurement scales determine which comparisons and numerical operations are meaningful.

Table 1.2: Measurement scales and the comparisons each scale supports.

| Scale    | What is meaningful?                                                                                | Examples / interpretation                                                                               |
|----------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Nominal  | Equality and inequality: $x_i = x_j$ or $x_i \neq x_j$ . No ordering is implied.                   | Hair color, eye color, gender, marital status.                                                          |
| Ordinal  | Equality plus order: $x_i < x_j$ or $x_i > x_j$ ; distances between categories are not quantified. | Educational qualification, military rank, job qualification.                                            |
| Interval | Ordered numerical scale with meaningful differences, but an arbitrary origin.                      | Celsius/Fahrenheit temperature, altitude. A 20° difference is meaningful, but 40°C is not “twice” 20°C. |
| Ratio    | Interval properties plus an intrinsic zero, making ratios meaningful.                              | Weight, height, income, counts of students or patients.                                                 |

### 1.3.1 Qualitative and quantitative variables

Qualitative variables may be nominal, ordinal, or binary/dichotomous. Quantitative variables may be **discrete**, when the values are finite or countable, or **continuous**, when the theoretical set of possible values is continuous. In practice, measurement precision discretizes even continuous phenomena: height recorded to the nearest centimetre or to two decimal places is a practical example.

Binary variables remain qualitative even when coded numerically as 0/1, 1/2, and so on. Similarly, an ordinal category can sometimes be converted to a numerical proxy, for example years of schooling corresponding to educational attainment; such transformations introduce conventions that can change the measurement assumptions.

#### Warning: Ordinal variables

A methodological difficulty arises when statistical procedures designed for interval-scale variables are often applied to ordinal variables even though, strictly speaking, ordinal data do not guarantee equal distances between categories. The scale assumptions of a method should therefore be considered explicitly.

Education levels can be harmonized across countries using an international classification such as ISCED. More generally, the number of categories may be fixed by the phenomenon (for example a binary variable) or chosen by the analyst (for example grouped height or salary intervals).

### 1.3.2 Cross-sectional and longitudinal studies

A **longitudinal study** repeatedly records variables on the same subjects over time. A **cross-sectional study** records variables on different subjects at a particular time point. The difference is structural: the former contains within-unit temporal information; the latter provides a snapshot across units.

## 1.4 Summation Notation

The capital Greek letter  $\Sigma$  is used throughout statistics to represent sums. For values  $x_1, \dots, x_n$ ,

$$\sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n. \quad (1.2)$$

$i$  dummy index identifying the current term;

$n$  number of terms;

$x_i$  the  $i$ th numerical observation.

For the interview-duration example 13, 15, 17, 12, 20 minutes,

$$\sum_{i=1}^5 x_i = 13 + 15 + 17 + 12 + 20 = 77.$$

With two indices, a double sum compactly represents observations indexed by two characteristics:

$$\sum_{i=1}^m \sum_{j=1}^n x_{ij}. \quad (1.3)$$

Here  $i$  and  $j$  identify the two indexing dimensions and  $x_{ij}$  is the observation at their intersection.

Useful summation identities include

$$\sum_{i=1}^n x_i^2 = x_1^2 + \dots + x_n^2, \quad \left( \sum_{i=1}^n x_i \right)^2 = (x_1 + \dots + x_n)^2,$$

and, for a constant  $a$ ,

$$\sum_{i=1}^n ax_i = a \sum_{i=1}^n x_i. \quad (1.4)$$

## 1.5 Frequency Distributions

### Definition: Absolute frequency

The absolute frequency  $n_i$  of category  $x_i$  is the number of statistical units that exhibit that category. If there are  $k$  categories and  $N$  total units,

$$\sum_{i=1}^k n_i = N. \quad (1.5)$$



The pairs  $(x_i, n_i)$  form an absolute frequency distribution. Data counting therefore assigns each observed unit to one and only one category for the variable under study. When several variables are present, the operation is repeated variable by variable; category assignment must be unambiguous.

### 1.5.1 Relative and percentage frequencies

The relative frequency and percentage frequency of category  $i$  are

$$f_i = \frac{n_i}{N}, \quad (1.6)$$

$$p_i = 100f_i = \frac{n_i}{N} \times 100. \quad (1.7)$$

|       |                                          |
|-------|------------------------------------------|
| $n_i$ | absolute frequency of category $i$ ;     |
| $N$   | total number of units;                   |
| $f_i$ | relative frequency, between 0 and 1;     |
| $p_i$ | percentage frequency, between 0 and 100. |

Consequently,

$$0 \leq f_i \leq 1, \quad \sum_{i=1}^k f_i = 1,$$

and

$$0 \leq p_i \leq 100, \quad \sum_{i=1}^k p_i = 100.$$

A residence-time example groups a quantitative variable into classes: 49 units have seven years or less, 5 lie in the middle class, and 46 have more than 16 years. The corresponding relative frequencies are 0.49, 0.05, and 0.46, and the percentage frequencies are 49%, 5%, and 46%.

### 1.5.2 Approximation and rounding

Rounding replaces an exact value by an approximation at a specified decimal precision. The difference introduced by the approximation is the rounding error. Examples include  $10.32 \rightarrow 10.3$  to one decimal place and  $10.7457 \rightarrow 10.746$  to three decimal places. Rounding can alter the apparent total of a frequency table slightly, so totals should be checked after reporting rounded relative or percentage frequencies.

### 1.5.3 Cumulative frequencies

For at least ordinal variables, categories can be ordered and cumulative frequencies can be formed. The cumulative absolute frequency at category  $x_i$  is

$$N_i = \sum_{j=1}^i n_j. \quad (1.8)$$

|       |                                                                  |
|-------|------------------------------------------------------------------|
| $N_i$ | number of observations in categories up to and including $x_i$ ; |
| $n_j$ | absolute frequency of category $j$ ;                             |
| $i$   | position of the current ordered category.                        |

The final cumulative absolute frequency is  $N_k = N$ . Cumulative relative and percentage frequencies are constructed analogously.

### 1.5.4 Grouping continuous values into classes

The salary example shows that grouping choices influence the apparent distribution. For 50 employees, one grouping is

Table 1.3: Salary frequencies for 50 employees under a four-class grouping.

| Salary class          | Absolute | Relative | Percentage |
|-----------------------|----------|----------|------------|
| $x \leq 4000$         | 23       | 0.46     | 46%        |
| $4000 < x \leq 6000$  | 7        | 0.14     | 14%        |
| $6000 < x \leq 10000$ | 13       | 0.26     | 26%        |
| $x > 10000$           | 7        | 0.14     | 14%        |
| Total                 | 50       | 1.00     | 100%       |

The same 50 observations can be represented with narrower classes, producing a more detailed distribution. The conceptual lesson is that class boundaries and widths are part of the analytical representation: changing them can change what patterns are visually prominent without changing the underlying raw observations.

## 1.6 Key Terms and Quick Review

| Term                                 | Meaning                                                                  |
|--------------------------------------|--------------------------------------------------------------------------|
| Statistical unit                     | Object on which variables are observed.                                  |
| Variable                             | Characteristic recorded on statistical units.                            |
| Category / modality                  | Possible manifestation of a variable.                                    |
| Nominal / ordinal / interval / ratio | Four levels of measurement with increasingly strong numerical structure. |
| Discrete / continuous                | Countable versus theoretically continuous quantitative values.           |
| Absolute frequency $n_i$             | Number of units in category $i$ .                                        |
| Relative frequency $f_i$             | $n_i/N$ .                                                                |
| Percentage frequency $p_i$           | $100n_i/N$ .                                                             |
| Cumulative frequency                 | Frequency accumulated up to an ordered category.                         |

### Takeaway: Quick review

A statistical analysis begins by defining the unit, the variables and their measurement scales. Those choices determine what operations are meaningful. Frequency distributions compress the data matrix into interpretable counts or proportions, while cumulative frequencies require an ordering of categories. These foundations determine which summaries and models are legitimate for a given dataset.

# Measures of Location and Data Visualization

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Measures of location describe representative values; graphs reveal the shape and structure that a single number cannot capture. This chapter connects mode, mean and median with graphical choices appropriate to each variable type.

## 2.1 From Data to Synthetic Measures

A population parameter is denoted generically by  $\theta$ . Frequency distributions and graphs provide structured descriptions of the data; measures of location go one step further by representing a distribution through a small number of characteristic values. This compression inevitably loses information, so a good summary should preserve the most relevant feature for the task.

### Definition: Measure of position

A measure of position or location is a synthetic value that describes where a distribution is centered or where its observations tend to concentrate.

## 2.2 Mode, Mean and Median

### 2.2.1 Mode

The **mode** is the category or value with the highest frequency. It can be defined for nominal, ordinal and quantitative variables. In an example on the number of dependent children under age 19, the most frequent value is 0.

The mode is most informative when the population is large enough for frequency differences to be informative. If every observation in a tiny dataset is unique, naming a mode adds little information.

### 2.2.2 Arithmetic mean

The arithmetic mean is defined for quantitative data by

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i. \quad (2.1)$$

$x_i$  value observed on unit  $i$ ;  
 $N$  number of observations;  
 $\mu$  population arithmetic mean.

It follows directly that

$$\sum_{i=1}^N (x_i - \mu) = 0, \quad (2.2)$$

A second central property is that the arithmetic mean is the value that minimizes the sum of squared deviations,

$$\mu = \arg \min_a \sum_{i=1}^N (x_i - a)^2. \quad (2.3)$$

$a$  any candidate summary value;  
 $x_i - a$  deviation of observation  $i$  from that candidate;  
 $\arg \min$  value of  $a$  producing the smallest squared-error loss.

For an absolute frequency distribution,

$$\mu = \frac{1}{N} \sum_{i=1}^k x_i n_i, \quad N = \sum_{i=1}^k n_i. \quad (2.4)$$

With relative and percentage frequencies,

$$\mu = \sum_{i=1}^k x_i f_i, \quad \mu = \frac{1}{100} \sum_{i=1}^k x_i p_i. \quad (2.5)$$

The frequencies therefore act as weights.

### 2.2.3 Weighted arithmetic mean

For nonnegative weights  $w_i$  with a positive total,

$$\mu_w = \frac{\sum_{i=1}^N x_i w_i}{\sum_{i=1}^N w_i}. \quad (2.6)$$

If the weights sum to one, this reduces to  $\mu_w = \sum_i x_i w_i$ . Relative frequencies are a specific instance of this weighting scheme.

### 2.2.4 Median

The median summarizes the center through the order of observations. It is meaningful for ordinal and quantitative variables even when numerical averaging is not. Detailed conventions and frequency-table examples appear in [section 3.1](#).

#### Example: Income and an extreme observation

For the ordered incomes

11,200, 11,400, 11,700, 12,200, 12,300, 12,500, 215,000,

the median is 12,200 because it occupies the fourth position. The extreme value has no effect on the median's position, a robustness property developed further in the next chapter.

## 2.3 Why Visualize Data?

Visualization summarizes data beyond the raw data matrix and frequency table. Visual representations exploit human pattern recognition and can reveal structure that is difficult to see in a table. The appropriate graph depends on the variable type, the analytical purpose, whether time or space is involved, and how many categories or series must be shown.

### 2.3.1 Pie charts

For a category with relative frequency  $f_i$  or percentage frequency  $p_i$ , the angle of the pie slice is

$$\theta_i = 360^\circ f_i = 3.6^\circ p_i. \quad (2.7)$$

|            |                                      |
|------------|--------------------------------------|
| $f_i$      | relative frequency in $[0, 1]$ ;     |
| $p_i$      | percentage frequency in $[0, 100]$ ; |
| $\theta_i$ | angle assigned to the category.      |

A relative frequency is a proportion, whereas a percentage is expressed on a scale from 0 to 100. The factor multiplying the frequency must match that convention.

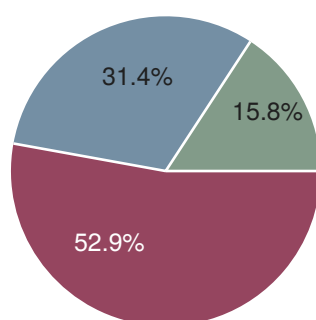


Figure 2.1: A three-category pie chart showing shares of 15.8%, 31.4% and 52.9%. Rounded labels sum to 100.1%; sector angles are normalized to a full circle.

Pie charts work best with few categories and clearly distinguishable shares. When categories are numerous or several shares are very small, bars generally provide clearer comparisons.

### 2.3.2 Bar charts, stick plots and multiple variables

Bar charts encode frequency through bar length or height and can be arranged horizontally or vertically. They remain useful for qualitative categories, but ordering should respect the natural order for ordinal variables.

For discrete quantitative variables, a useful representation is a **stick plot**: values remain at their correct numerical positions on the horizontal axis and frequency is represented by vertical rods rather than adjacent bars. Grouped bar charts can show more than one variable or group simultaneously when comparisons remain readable.

### 2.3.3 Time series and scale choice

Time-series plots display values against time and are effective for showing trend. Axis scaling can change the perceived magnitude of variation. A narrow vertical range visually magnifies small movements; a wide range compresses them.

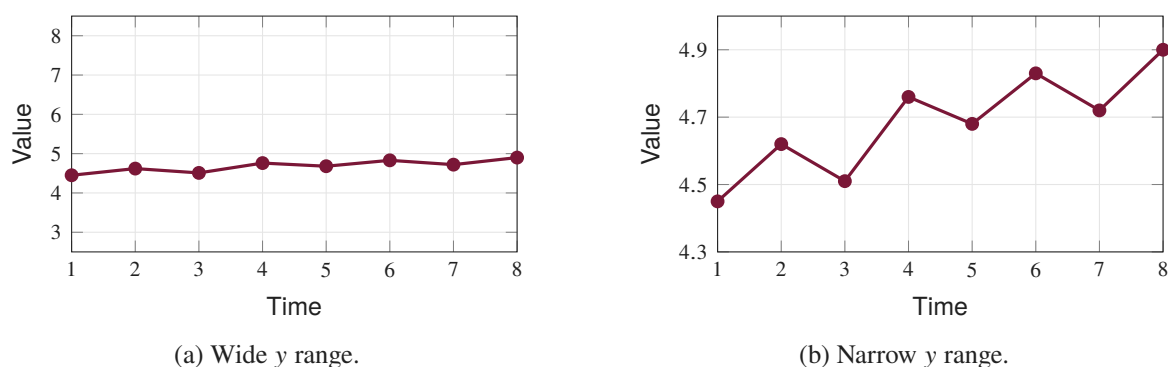


Figure 2.2: Schematic effect of vertical-axis scaling: identical values can look much more volatile when the vertical axis is truncated. Numerical coordinates are illustrative.

When many time series must be compared, small multiples place each series on its own panel with consistent axes, avoiding dense bundles of overlapping lines.

### 2.3.4 Maps

A map can encode a variable spatially when geography is part of the phenomenon. For example, a map can show the percentage of smokers across areas of the East of England. As with every visualization, the map is useful only when the spatial encoding serves the analytical question rather than merely decorating the data.

## 2.4 Histograms and Unequal Class Widths

Histograms are appropriate for continuous quantitative variables grouped into intervals. Their defining feature is that the *area* of a bar, not merely its height, represents frequency. Therefore unequal class widths require density-adjusted heights.

A leaf-length example uses 1,900 leaf lengths of *Vaccinium vitis-idaea*, ranging from approximately 0 to 17.5 mm. When equal-width bins are used, plotting frequency as height preserves proportional area automatically. When class widths differ, density must be used.

For class  $i$  with width  $a_i$ , relative frequency  $f_i$  and absolute frequency  $n_i$ ,

$$h_i^{(\text{rel})} = \frac{f_i}{a_i}, \quad (2.8)$$

$$h_i^{(\text{abs})} = \frac{n_i}{a_i}. \quad (2.9)$$

|       |                                   |
|-------|-----------------------------------|
| $a_i$ | width of class $i$ ;              |
| $f_i$ | relative frequency of class $i$ ; |
| $n_i$ | absolute frequency of class $i$ ; |
| $h_i$ | density used as bar height.       |

With unequal class widths, dividing frequency by width ensures that bar area represents frequency. Relative-frequency densities give a histogram of total area one.

The counts and densities for three unequal-width classes are:

Table 2.1: Unequal-width classes for the leaf-length example.

| Class (mm) | $n_i$ | $f_i$   | $a_i$ | $f_i/a_i$ |
|------------|-------|---------|-------|-----------|
| (0, 4]     | 431   | 0.22684 | 4     | 0.05671   |
| (4, 10]    | 1408  | 0.74105 | 6     | 0.12351   |
| (10, 20]   | 61    | 0.03211 | 10    | 0.00321   |
| Total      | 1900  | 1.00    |       |           |



Figure 2.3: Unequal-width histogram for 1,900 leaf-length observations. The visual area of each rectangle, not its height alone, encodes class frequency.

#### Takeaway: Quick review

The mean minimizes squared deviations but is sensitive to extremes; the median is order-based and the mode is frequency-based. Graphical choices depend on measurement scale: nominal variables use bars or pies, ordinal categories retain their order, and discrete quantitative values suit stick plots. Maps serve spatial questions, and continuous data grouped into intervals use histograms whose areas encode frequencies. Together, location measures and plots describe representative values and distribution shape. Correct scales, ordering and density are essential to interpretation.

# Median, Quantiles, Variability and Outliers

Distributions differ in spread and shape even when their centers coincide. Quantiles provide order-based summaries, variance measures dispersion around the mean, and outlier diagnostics identify observations requiring closer investigation.

## 3.1 The Median in Detail

Position-based summaries such as the median are especially useful for rankings, thresholds and robust descriptions. Examples include dividing football teams into performance groups or selecting the top 25% of students after ordering grades.

### Definition: Median

The median, denoted  $Me$  or  $x_{0.5}$ , is a value with at least half the observations at or below it and at least half at or above it. For numerical data with an even count, the usual convention selects the midpoint of the two central observations.

### 3.1.1 Odd sample size

For  $N$  odd, the median position is

$$P = \frac{N + 1}{2}. \quad (3.1)$$

For the temperatures 8, 10, 5, 3, 11, ordering gives 3, 5, 8, 10, 11, so  $P = 3$  and the median is 8. The mean is  $37/5 = 7.4$ .

For 13, 12, 7, 10, 8, the ordered sequence is 7, 8, 10, 12, 13 and the median is 10, equal to the mean  $50/5 = 10$ . The same position logic applies to ordinal data: *poor, poor, good, excellent, excellent* has median category *good*, even though a numerical mean is not meaningful.

A larger ordered example of 25 residence-time observations gives  $P = (25 + 1)/2 = 13$  and a median of 33 years; the mean is  $761/25 = 30.44$ .

### 3.1.2 Even sample size

For quantitative data with  $N$  even, the two central positions are

$$P_1 = \frac{N}{2}, \quad P_2 = \frac{N}{2} + 1, \quad (3.2)$$

and the common numerical convention is

$$x_{0.5} = \frac{x_{(P_1)} + x_{(P_2)}}{2}. \quad (3.3)$$



Here  $x_{(r)}$  denotes the observation at ordered position  $r$ .

Six-observation examples illustrate the numerical convention. For 3, 5, 8, 8, 10, 11, the two middle values are both 8, so the median is 8. For 3, 5, 7, 8, 10, 11, the median is  $(7 + 8)/2 = 7.5$ . For 7, 9, 11, 13, 15, 17, the median is 12, which equals the mean.

For ordinal data, averaging category labels is not meaningful. If the two middle ordered categories are the same (for example *good*, *good*) the median is that category. If they differ (for example *good*, *excellent*), arithmetic averaging cannot produce a meaningful median category; the two central categories describe the median interval.

### 3.1.3 Median from frequency tables

Cumulative frequencies locate the median without expanding every observation. For  $N = 123$  children-count observations, the median position is  $(123 + 1)/2 = 62$ ; because the first cumulative frequency is 75 at category 0, the median is 0.

For  $N = 124$ , the middle positions are 62 and 63. If both fall in category 0, the median is 0. If instead position 62 is in category 0 and position 63 in category 1, yielding the numerical median  $(0 + 1)/2 = 0.5$ .

For continuous data grouped into classes, the strategy is to compute cumulative frequencies, identify the median class, then locate  $x_{0.5}$  inside that class. The median class is identifiable from cumulative counts, but a numerical value within that class requires an assumption about the within-class distribution.

## 3.2 Mean, Median, Mode and Distribution Shape

For a symmetric unimodal distribution with a unique central mode and a finite mean,

$$Mo = x_{0.5} = \mu. \quad (3.4)$$

For many positively skewed (right-skewed) unimodal distributions, a common pattern is

$$Mo < x_{0.5} < \mu, \quad (3.5)$$

while the corresponding pattern for many negatively skewed (left-skewed) distributions is

$$Mo > x_{0.5} > \mu. \quad (3.6)$$

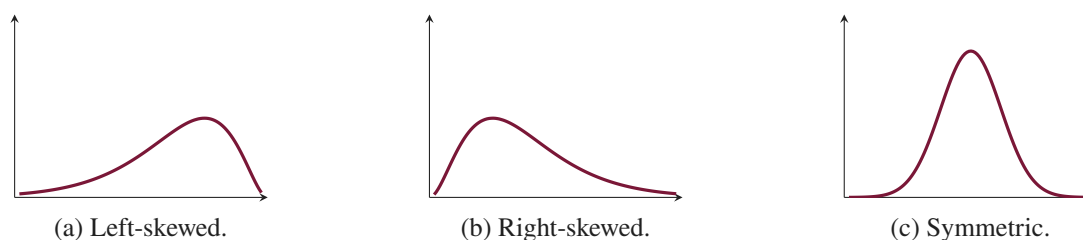


Figure 3.1: Schematic left-skewed, right-skewed and symmetric distribution shapes. Curves illustrate tail direction rather than empirical data.

The orderings in equations (3.5) and (3.6) are common patterns, not universal theorems about all skewed distributions.

The median is robust to extreme observations because it depends on order rather than magnitude. For example, adding one score of 30 to 23 scores of 18 leaves the median at 18 while the mean rises from 18 to 18.5. Conversely, the median is less algebraically convenient than the mean and is more difficult to manipulate in later inferential theory.

### 3.3 Quantiles and the Boxplot

Quantiles generalize the median. Quantiles of order  $n$  consist of  $n - 1$  cut points dividing an ordered distribution into  $n$  equal-probability parts. Quartiles use  $n = 4$ , deciles  $n = 10$ , and percentiles  $n = 100$ . The median is therefore the second quartile, fifth decile and 50th percentile. For a cumulative distribution function  $F$ , a quantile at probability  $q \in (0, 1)$  can be defined as  $x_q = \inf\{x : F(x) \geq q\}$ . Sample quantiles use conventions for selecting or interpolating ordered values; different conventions can give different results. In a discrete distribution, ties prevent exact division into equal-sized groups. Here the order  $n$  counts probability groups, rather than observations.

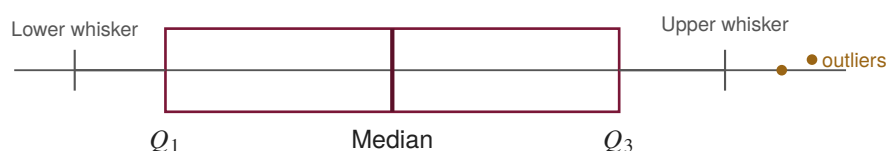


Figure 3.2: Schematic boxplot: the box spans the quartiles, the central line marks the median, and whiskers end at the most extreme observations within the fences.

Useful revision questions include: can there be more than one median, arithmetic mean, or mode; can there be more than one quartile; how many order- $n$  quantiles are there; can  $Q_1 = Q_3$ ; can  $Q_1 > Q_3$ ; is there a fourth quartile; and, with only 50 units, can one meaningfully compute all percentiles? These questions reinforce that quantiles are ordered cut points, whereas mode is a frequency property and can be multimodal. For nonempty numerical data, the arithmetic mean is unique, while several values may share the highest frequency. A median can be an interval of admissible values, although the midpoint convention selects one numerical value. Quartiles satisfy  $Q_1 \leq Q_3$  and can coincide; three interior cut points define four groups. Percentiles can be computed from 50 observations by interpolation, but they do not create additional information or 100 distinct empirical groups. In a standard boxplot, the fences are  $Q_1 - 1.5IQR$  and  $Q_3 + 1.5IQR$ . Whiskers end at the smallest and largest observed values within these fences; they generally do not end at the fence values themselves. The diagram is schematic and has no numerical axis.

### 3.4 Measures of Variability

A measure of center cannot describe spread. Two groups can have the same mean but radically different internal variation. A regional-income example compares incomes 1000, 1500, 2000 with three identical incomes of 1500: both means equal 1500, but only the first distribution varies.

### 3.4.1 Range and interquartile spread

The **range** is  $x_{\max} - x_{\min}$ . The interquartile range and semi-interquartile range are

$$IQR = Q_3 - Q_1, \quad (3.7)$$

$$SIQR = \frac{Q_3 - Q_1}{2}. \quad (3.8)$$

The interval from  $Q_1$  to  $Q_3$  describes the central half of the distribution; ties can place more than half the observations within it.

### 3.4.2 Variance

For a population of  $N$  observations,

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2. \quad (3.9)$$

|             |                                             |
|-------------|---------------------------------------------|
| $\mu$       | arithmetic mean;                            |
| $x_i - \mu$ | deviation of observation $i$ from the mean; |
| $N$         | number of observations;                     |
| $\sigma^2$  | population variance.                        |

For a frequency distribution,

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^k (x_i - \mu)^2 n_i = \sum_{i=1}^k (x_i - \mu)^2 f_i. \quad (3.10)$$

The numerator  $\sum_i (x_i - \mu)^2$  is the **sum of squared deviations**, also called the descriptive deviance here; it is distinct from likelihood-based model deviance. Larger variance indicates greater dispersion around the mean, measured in squared units.

### 3.4.3 Standard deviation

The standard deviation restores the original measurement unit:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}. \quad (3.11)$$

It satisfies  $\sigma \geq 0$ , equals zero only when all observations are identical, and increases as observations spread farther from the mean. For a sample of  $n > 1$  observations, the usual variance estimator is  $s^2 = (n - 1)^{-1} \sum_{i=1}^n (x_i - \bar{x})^2$ , where  $\bar{x}$  is the sample mean. The divisor  $n - 1$  adjusts for estimating the mean from the same sample.

### 3.4.4 Coefficient of variation

To compare variability across distributions whose means or scales differ, the coefficient of variation is

$$CV = \frac{\sigma}{\mu}. \quad (3.12)$$

|          |                                     |
|----------|-------------------------------------|
| $\sigma$ | standard deviation;                 |
| $\mu$    | mean;                               |
| $CV$     | scale-relative variability measure. |

The coefficient is meaningful for ratio-scale quantities with a positive mean. It is undefined at zero mean and unstable near zero; arbitrary-origin scales such as Celsius temperature do not support its usual interpretation. Multiplying by 100 expresses it as a percentage.

## 3.5 Outliers

Outliers matter because they may represent recording errors, genuinely exceptional units, or observations that strongly influence an analysis. Sensitivity can be checked by repeating an analysis with and without a suspicious observation, documenting the reason for any exclusion.

Two rules are presented:

- **IQR rule:** values below  $Q_1 - 1.5IQR$  or above  $Q_3 + 1.5IQR$  are potential outliers;
- **z-score rule:** observations more than roughly three standard deviations from the mean are potential outliers.

The standardized score is

$$z_i = \frac{x_i - \mu}{\sigma}. \quad (3.13)$$

|          |                                                                      |
|----------|----------------------------------------------------------------------|
| $x_i$    | observation under examination;                                       |
| $\mu$    | mean;                                                                |
| $\sigma$ | standard deviation;                                                  |
| $z_i$    | signed number of standard deviations separating $x_i$ from the mean. |

This score requires  $\sigma > 0$ . The three-standard-deviation rule is a screening convention whose suitability depends on distribution shape.

#### Worked Example: The seven-income example

**Given.** The incomes are 11,200, 11,400, 11,700, 12,200, 12,300, 12,500, and 215,000 euro.

**Procedure.** Using the population formulas gives  $\mu = 40,900$  and  $\sigma \approx 71,077.42$ . The final income has  $z \approx 2.45$ .

**Result.** Under the specific “ $|z| > 3$ ” rule, the extreme value is not flagged, despite being visually exceptional.

**Interpretation.** This illustrates why outlier detection depends on the rule: an extreme observation can inflate both the mean and standard deviation and thereby reduce its own standardized distance. The IQR criterion is much more robust in such a case.

### 3.6 Variability of Qualitative Variables: Gini Impurity

For qualitative variables, variability concerns heterogeneity across categories rather than numerical spread. A heterogeneity measure should be nonnegative, minimized when all units occupy one category, and maximized when frequencies are evenly distributed.

#### Definition: Gini impurity index

For relative category frequencies  $f_1, \dots, f_k$ ,

$$E_1 = 1 - \sum_{i=1}^k f_i^2. \quad (3.14)$$

The maximum is  $(k - 1)/k$ , attained when  $f_i = 1/k$  for every category.

The normalized form is

$$E_1^* = E_1 \frac{k}{k - 1}, \quad (3.15)$$

which ranges from 0 to 1 when  $k > 1$ .

At equal category frequencies,

$$E_{1,\max} = 1 - k \left( \frac{1}{k} \right)^2 = 1 - \frac{1}{k} = \frac{k - 1}{k}.$$

In the four-category university example, the squared relative frequencies sum to 0.2636, so

$$E_1 = 1 - 0.2636 = 0.7364, \quad E_1^* = 0.7364 \cdot \frac{4}{3} \approx 0.9819.$$

The normalized value is close to one, indicating strong heterogeneity across the four categories.

#### Takeaway: Quick review

The median and quantiles depend on order; variance and standard deviation depend on squared deviations from the mean; the coefficient of variation expresses spread relative to the mean; and Gini impurity expresses heterogeneity in categorical distributions. No single summary is sufficient for every distribution, especially in the presence of skewness or outliers.

# Bivariate Analysis, Association and Simple Regression

Bivariate analysis studies the joint behavior of two variables. Two-way tables describe categorical association, covariance and correlation quantify quantitative relationships, and regression models a linear prediction from one variable to another.

## 4.1 From One Variable to Two

Bivariate analysis observes two variables  $X$  and  $Y$  on the same  $N$  statistical units, producing paired observations

$$(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N). \quad (4.1)$$

Two-way tables summarize such pairs when variables are categorical or discretized into categories.

## 4.2 Joint and Marginal Frequencies

Let  $X$  have  $h$  categories  $x_1, \dots, x_h$  and  $Y$  have  $k$  categories  $y_1, \dots, y_k$ .

**Definition: Joint absolute frequency**

$n_{ij}$  is the number of units simultaneously exhibiting  $X = x_i$  and  $Y = y_j$ .

The corresponding joint relative frequency is

$$f_{ij} = \frac{n_{ij}}{N}. \quad (4.2)$$

Marginal totals are obtained by summing over the other variable:

$$n_{i\cdot} = \sum_{j=1}^k n_{ij}, \quad n_{\cdot j} = \sum_{i=1}^h n_{ij}. \quad (4.3)$$

The analogous relative marginals are  $f_{i\cdot} = n_{i\cdot}/N$  and  $f_{\cdot j} = n_{\cdot j}/N$ .

A cross-tabulation of attachment to the European Union by gender illustrates joint and marginal counts.

Dividing each count by its gender total gives percentages within gender. For men the five percentages are 9.7%, 37.3%, 35.3%, 16.3%, and 1.4%; for women they are 8.1%, 34.1%, 38.0%, 17.5%, and 2.3%. Overall, the corresponding shares are 8.8%, 35.6%, 36.8%, 17.0%, and 1.9%. This paired presentation distinguishes joint cell counts from conditional percentage profiles.

Table 4.1: Generic two-way table of joint absolute frequencies.

|          | $y_1$         | $y_2$         | $\cdots$ | $y_k$         | Total        |
|----------|---------------|---------------|----------|---------------|--------------|
| $x_1$    | $n_{11}$      | $n_{12}$      | $\cdots$ | $n_{1k}$      | $n_{1\cdot}$ |
| $x_2$    | $n_{21}$      | $n_{22}$      | $\cdots$ | $n_{2k}$      | $n_{2\cdot}$ |
| $\vdots$ | $\vdots$      | $\vdots$      | $\ddots$ | $\vdots$      | $\vdots$     |
| $x_h$    | $n_{h1}$      | $n_{h2}$      | $\cdots$ | $n_{hk}$      | $n_{h\cdot}$ |
| Total    | $n_{\cdot 1}$ | $n_{\cdot 2}$ | $\cdots$ | $n_{\cdot k}$ | $N$          |

Table 4.2: Attachment to the European Union by gender: joint counts and marginal totals. DK denotes “don’t know”.

| Attachment to the EU | Man    | Woman  | Total  |
|----------------------|--------|--------|--------|
| Very attached        | 1,238  | 1,212  | 2,450  |
| Fairly attached      | 4,769  | 5,128  | 9,897  |
| Not very attached    | 4,511  | 5,722  | 10,233 |
| Not at all attached  | 2,088  | 2,633  | 4,721  |
| DK                   | 182    | 346    | 528    |
| Total                | 12,788 | 15,041 | 27,829 |

### 4.3 Statistical Independence

If  $Y$  behaves the same conditionally on every category of  $X$  as it does marginally, then the variables are statistically independent. In frequency form,

$$\frac{n_{ij}}{n_{i\cdot}} = \frac{n_{\cdot j}}{N} \quad \text{for all } i, j. \quad (4.4)$$

Equivalently,

$$n_{ij}^* = \frac{n_{i\cdot} n_{\cdot j}}{N}, \quad (4.5)$$

where  $n_{ij}^*$  is the theoretical or expected frequency under independence.

|               |                                                          |
|---------------|----------------------------------------------------------|
| $n_{ij}$      | observed joint frequency;                                |
| $n_{i\cdot}$  | marginal frequency for category $x_i$ ;                  |
| $n_{\cdot j}$ | marginal frequency for category $y_j$ ;                  |
| $N$           | total number of units;                                   |
| $n_{ij}^*$    | expected joint frequency if $X$ and $Y$ are independent. |

The difference

$$c_{ij} = n_{ij} - n_{ij}^* \quad (4.6)$$

is the observed-minus-expected cell discrepancy, or **contingency**. Dependence can therefore be measured by aggregating the discrepancies between observed and independence-expected cells.

### 4.3.1 Pearson's $\chi^2$ index

$$\chi^2 = \sum_{i=1}^h \sum_{j=1}^k \frac{(n_{ij} - n_{ij}^*)^2}{n_{ij}^*} = \sum_{i=1}^h \sum_{j=1}^k \frac{c_{ij}^2}{n_{ij}^*}. \quad (4.7)$$

|          |                                                      |
|----------|------------------------------------------------------|
| $h, k$   | number of categories of $X$ and $Y$ ;                |
| $c_{ij}$ | observed-minus-expected contingency;                 |
| $\chi^2$ | nonnegative aggregate discrepancy from independence. |

The index is zero exactly when all observed cell frequencies equal their expected independent frequencies; it increases as discrepancies become larger. Remove categories with zero marginal totals so that all expected counts are positive. As a descriptive index,  $\chi^2$  also depends on dataset size. Its use as a hypothesis-test statistic requires additional sampling assumptions.

A normalized version is

$$\chi_{\text{norm}}^2 = \frac{\chi^2}{N \min\{h-1, k-1\}}, \quad (4.8)$$

which lies between 0 and 1 for  $h, k > 1$  after removing empty marginal categories. It is the square of Cramér's  $V$ .

## 4.4 Covariance and Correlation

For quantitative variables, covariance measures joint variation:

$$\sigma_{xy} = \frac{1}{N} \sum_{i=1}^N (x_i - \mu_x)(y_i - \mu_y). \quad (4.9)$$

|                |                                  |
|----------------|----------------------------------|
| $\mu_x, \mu_y$ | means of $X$ and $Y$ ;           |
| $\sigma_{xy}$  | covariance between $X$ and $Y$ ; |
| $N$            | number of paired observations.   |

A positive covariance indicates that above-average values tend to occur together; a negative covariance indicates opposite-signed deviations. Its magnitude depends on the measurement units, motivating standardization.

The Pearson linear correlation coefficient is

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}, \quad -1 \leq \rho_{xy} \leq 1. \quad (4.10)$$

Positive values indicate positive linear association, negative values indicate negative linear association, and zero indicates no *linear* correlation. The coefficient requires  $\sigma_x, \sigma_y > 0$ . Its absolute magnitude measures linear strength, with  $|\rho| = 1$  for an exact affine relationship between nonconstant variables. Association alone does not establish causation.

#### Key Concept: Zero correlation is not “no relationship”

A dataset can have correlation near zero while exhibiting a strong nonlinear pattern. Pearson correlation measures linear association; nonlinear dependence can remain when  $\rho = 0$ .



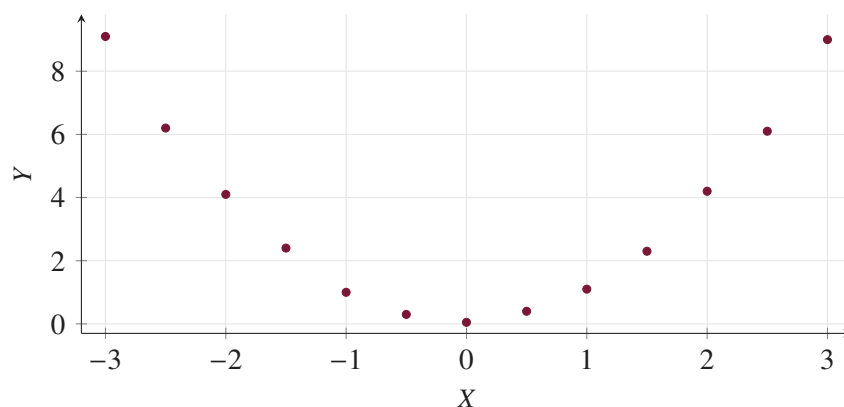


Figure 4.1: Schematic nonlinear relationship: a strong pattern can coexist with approximately zero linear correlation. Coordinates are illustrative.

#### 4.4.1 Spearman rank correlation

Spearman correlation measures monotone association through ranks. Assign ranks to  $X$  and  $Y$ . When there are no ties, the coefficient can be computed from rank differences  $d_i$  as

$$\rho_S = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}. \quad (4.11)$$

The shortcut in equation (4.11) applies only when there are no ties. With ties, assign average ranks and compute Pearson correlation on those ranks. Here  $d_i$  is the difference between the two ranks for observation  $i$ , and  $n > 1$  is the number of pairs; both rank vectors must have positive variance.

### 4.5 Scatterplots and Types of Relationship

Scatterplots are used to diagnose positive linear, negative linear, nonlinear and absent-looking relationships. Repeated discrete  $x$  values can create vertical bands of points. Graphical inspection complements, rather than replaces, numerical association measures.

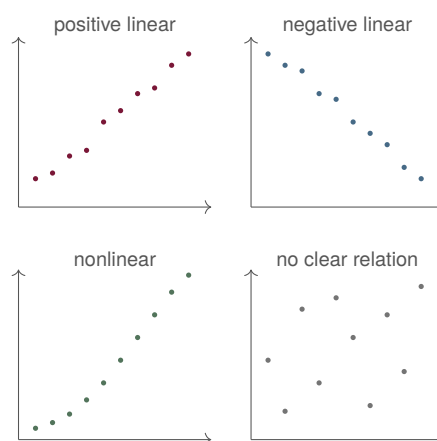


Figure 4.2: Schematic scatterplots distinguish positive and negative linear association, nonlinear association and the absence of a clear pattern.

## 4.6 Ordinary Least Squares Regression

A simple linear prediction takes the form

$$\hat{y}_i = a + bx_i, \quad (4.12)$$

and chooses  $a$  and  $b$  by minimizing the sum of squared residuals:

$$\min_{a,b} \sum_{i=1}^N (y_i - a - bx_i)^2. \quad (4.13)$$

|                   |                                    |
|-------------------|------------------------------------|
| $a$               | intercept;                         |
| $b$               | slope or regression coefficient;   |
| $\hat{y}_i$       | fitted value for observation $i$ ; |
| $y_i - \hat{y}_i$ | residual;                          |
| $N$               | number of paired observations.     |

Using descriptive covariance and variance computed with the same divisor,

$$b = \frac{\sigma_{xy}}{\sigma_x^2}, \quad a = \mu_y - b\mu_x. \quad (4.14)$$

The relationship to Pearson correlation is

$$b = \rho_{xy} \frac{\sigma_y}{\sigma_x}, \quad \rho_{xy} = b \frac{\sigma_x}{\sigma_y}. \quad (4.15)$$

Thus slope and correlation share the direction of linear association, while slope additionally depends on the scales of  $X$  and  $Y$ . The slope requires  $\sigma_x^2 > 0$  and represents the change in fitted  $Y$  per unit increase in  $X$ . Using covariance and variance with the common sample divisor  $n - 1$  gives the same slope; the intercept then uses  $\bar{y} - b\bar{x}$ .

### Takeaway: Quick review

Two-way tables describe joint distributions; independence compares conditional behavior with marginal behavior. Pearson's  $\chi^2$  summarizes observed-versus-expected discrepancies for categorical variables. Covariance and Pearson correlation describe quantitative co-variation, Spearman correlation uses ranks for monotone association, and ordinary least squares chooses the line minimizing squared vertical residuals.

# Multivariate Analysis and Linear Algebra Foundations

---

A multivariate dataset records several variables for each unit. Matrix operations make its structure explicit and provide the foundations for covariance matrices, transformations and decompositions used in statistical analysis.

## 5.1 Why Linear Algebra Appears in Multivariate Statistics

Multivariate analysis studies the structure of a data matrix containing several variables simultaneously. A multivariate dataset may contain characteristics such as sex, race, height, income, marital status, years of education and a liberal–conservative score. A Zenodo record (<https://zenodo.org/records/16947276>) contains the example file `sample_trips_mvgrad_june2023.csv`.

## 5.2 Matrices, Vectors and Indexing

A matrix is a rectangular array of elements arranged in rows and columns. For example,

$$A = \begin{bmatrix} 1000 & 23 \\ 3000 & 47 \\ 2000 & 35 \end{bmatrix}$$

has dimension  $3 \times 2$ . In general,  $A = [a_{ij}]$ , where  $i$  indexes rows and  $j$  indexes columns.

A square matrix has the same number of rows and columns. A column vector is an  $n \times 1$  matrix; a row vector is a  $1 \times n$  matrix; and a scalar can be viewed as a  $1 \times 1$  matrix.

### 5.2.1 Transpose and equality

The transpose exchanges rows and columns:

$$A = \begin{bmatrix} 2 & 5 \\ 7 & 10 \\ 3 & 4 \end{bmatrix} \implies A^T = \begin{bmatrix} 2 & 7 & 3 \\ 5 & 10 & 4 \end{bmatrix}.$$

Two matrices are equal when their dimensions and all corresponding entries are equal.

### 5.3 Matrix Operations

Addition and subtraction operate element by element on matrices of the same dimensions. Multiplication by a scalar multiplies every entry.

Matrix multiplication is dimension-sensitive. If  $A$  is  $n \times p$  and  $B$  is  $p \times k$ , then  $AB$  exists and is  $n \times k$ . For example,

$$A = \begin{bmatrix} 1 & 3 & 4 \\ 0 & 5 & 8 \end{bmatrix}, \quad B = \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix}, \quad AB = \begin{bmatrix} 26 \\ 41 \end{bmatrix}.$$

### 5.4 Special Matrices

A square matrix is **symmetric** when  $A = A^T$ . A **diagonal matrix** has zeros off the main diagonal. The **identity matrix**  $I$  has ones on the diagonal and satisfies

$$AI_p = I_n A = A \quad (A \in \mathbb{R}^{n \times p}). \quad (5.1)$$

Let  $\mathbf{1}$  denote the  $n$ -dimensional all-ones column vector and  $J$  the  $n \times n$  all-ones matrix. Then

$$\mathbf{1}\mathbf{1}^T = J. \quad (5.2)$$

### 5.5 Linear Dependence and Rank

#### Definition: Linear dependence

Columns  $C_1, \dots, C_c$  are linearly dependent if scalars  $k_1, \dots, k_c$ , not all zero, exist such that

$$k_1 C_1 + \dots + k_c C_c = \mathbf{0}. \quad (5.3)$$

If the only solution is  $k_1 = \dots = k_c = 0$ , the columns are linearly independent.

The matrix

$$\begin{bmatrix} 1 & 2 & 5 & 1 \\ 2 & 2 & 10 & 6 \\ 3 & 4 & 15 & 1 \end{bmatrix}$$

has a third column equal to five times the first, so its columns are linearly dependent.

#### Definition: Rank

The rank of a matrix is the maximum number of linearly independent columns; equivalently, it is the maximum number of linearly independent rows.

For an  $r \times c$  matrix,  $\text{rank}(A) \leq \min(r, c)$ . Multiplying by nonsingular matrices on the left or right preserves rank when dimensions are compatible.

### 5.6 Inverse Matrices and Determinants

For an invertible square matrix,

$$AA^{-1} = A^{-1}A = I. \quad (5.4)$$

Not every nonzero matrix is invertible. For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

if  $ad - bc \neq 0$ ,

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}. \quad (5.5)$$

The quantity  $ad - bc$  is the determinant. A square matrix is nonsingular exactly when it has full rank and nonzero determinant.

For example,

$$A = \begin{bmatrix} 2 & 4 \\ 3 & 1 \end{bmatrix}, \quad A^{-1} = \begin{bmatrix} -0.1 & 0.4 \\ 0.3 & -0.2 \end{bmatrix}$$

indeed satisfies  $A^{-1}A = I$ .

### 5.6.1 Minors and cofactors

For  $A = [a_{ij}]$ , let  $M_{ij}$  be the submatrix formed by deleting row  $i$  and column  $j$ . The cofactor is

$$A_{ij} = (-1)^{i+j} \det(M_{ij}). \quad (5.6)$$

Cofactor expansion along row  $i$  is

$$\det(A) = \sum_{j=1}^n a_{ij} A_{ij}, \quad (5.7)$$

and the analogous expansion can be performed down any column.

Applied to

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix},$$

the first-row minors are  $-3, -6, -3$ , the cofactors are  $-3, 6, -3$ , and the determinant is  $-3 + 12 - 9 = 0$ . Therefore the matrix is singular.

### 5.6.2 Using an inverse to solve a system

The ordinary equation  $5y = 20$  is solved by multiplying by  $1/5$ . In matrix notation, if

$$AY = C, \quad (5.8)$$

and  $A$  is invertible, then

$$Y = A^{-1}C. \quad (5.9)$$

An example is the system  $2y_1 + 4y_2 = 20$  and  $3y_1 + y_2 = 10$ . With the inverse above and  $C = (20, 10)^T$ , multiplication gives  $Y = (2, 4)^T$ ; substitution verifies both equations.

## 5.7 Eigenvalues and Eigenvectors

### Definition: Eigenpair

Let  $x \neq \mathbf{0}$  and let  $M$  be an  $n \times n$  matrix. If

$$Mx = \lambda x, \quad (5.10)$$

then  $\lambda$  is an eigenvalue of  $M$  and  $x$  is a corresponding eigenvector.

The eigenvalues satisfy the characteristic equation

$$\det(M - \lambda I) = 0, \quad (5.11)$$

which is equivalent, up to an overall sign convention, to  $\det(\lambda I - M) = 0$ .

For a column eigenvector, the equivalent homogeneous system is  $(M - \lambda I)x = \mathbf{0}$ . Matrix multiplication must preserve this order.

A square matrix is nonsingular if and only if zero is not one of its eigenvalues. For a matrix with characteristic factorization

$$\det(A - \lambda I) = (3 - \lambda)(-\lambda)(2 - \lambda),$$

the eigenvalues are 3, 0 and 2. The zero eigenvalue establishes singularity.

## 5.8 Orthogonal Matrices and Singular Value Decomposition

A real square matrix  $A$  is orthogonal when

$$AA^T = I. \quad (5.12)$$

Any real  $m \times n$  matrix admits a singular value decomposition

$$A = U\Sigma V^T, \quad (5.13)$$

Here  $U \in \mathbb{R}^{m \times m}$  and  $V \in \mathbb{R}^{n \times n}$  are orthogonal, and  $\Sigma \in \mathbb{R}^{m \times n}$  is rectangular diagonal with nonnegative singular values. Its nonzero singular values count the rank of  $A$ . Here  $\Sigma$  is a singular-value matrix; in covariance formulas the same Greek letter denotes a different object.

For a real symmetric matrix, the singular values are the absolute eigenvalue magnitudes  $|\lambda_i|$ , arranged in nonincreasing order. A negative eigenvalue therefore contributes a positive singular value of the same magnitude.

## 5.9 Useful Matrix Identities

The following identities hold when the dimensions are compatible and the indicated inverses exist:

$$\begin{aligned}
 A + B &= B + A, & (A + B) + C &= A + (B + C), \\
 (AB)C &= A(BC), & C(A + B) &= CA + CB, \\
 k(A + B) &= kA + kB, & (A^\top)^\top &= A, \\
 (A + B)^\top &= A^\top + B^\top, & (AB)^\top &= B^\top A^\top, \\
 (ABC)^\top &= C^\top B^\top A^\top, & (AB)^{-1} &= B^{-1}A^{-1}, \\
 (ABC)^{-1} &= C^{-1}B^{-1}A^{-1}, & (A^{-1})^{-1} &= A, \\
 (A^\top)^{-1} &= (A^{-1})^\top.
 \end{aligned}$$

## 5.10 Variance–Covariance and Correlation Matrices

Let  $X$  be an  $n \times p$  data matrix with variable columns  $X_1, \dots, X_p$ . The covariance matrix is

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \cdots & \sigma_p^2 \end{bmatrix}. \quad (5.14)$$

The diagonal contains variances; off-diagonal entries contain covariances, and  $\sigma_{ij} = \sigma_{ji}$ .

The correlation matrix standardizes these covariances:

$$R = \begin{bmatrix} 1 & r_{12} & \cdots & r_{1p} \\ r_{21} & 1 & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p1} & r_{p2} & \cdots & 1 \end{bmatrix}, \quad r_{ij} = \frac{\sigma_{ij}}{\sigma_i \sigma_j}. \quad (5.15)$$

Both matrices are symmetric and positive semidefinite. Under the descriptive population convention,  $\sigma_{ij} = n^{-1} \sum_{\ell=1}^n (x_{\ell i} - \mu_i)(x_{\ell j} - \mu_j)$ ; using the divisor  $n - 1$  and sample means gives sample covariances. Correlation entries require positive component variances. Here  $r_{ij}$  denotes a standardized matrix entry; it equals  $\rho_{ij}$  under the population convention. These matrices condense the pairwise relationships among all  $p$  variables.

### Takeaway: Quick review

Linear algebra provides a compact language for multivariate data. Rank describes independent dimensional structure; invertibility is tied to full rank, nonzero determinant and absence of a zero eigenvalue; SVD factorizes any real rectangular matrix; covariance and correlation matrices summarize pairwise relationships among all variables at once.

# R for Statistical Computing

---

R connects statistical methods with executable calculations, data manipulation and graphical display. This chapter introduces the language, its historical background, the distinction between R and RStudio, and the basic ways to run commands. Scripts, data structures, operations, control flow and functions provide the wider programming framework for statistical applications.

## 6.1 What R Is

R is an integrated suite for data manipulation, calculation and graphical display, and a programming language supporting procedural programming with functions as well as object-oriented programming through generic functions. It is free and open source, flexible, and strongly oriented toward statistical computing.

The package ecosystem extends the core language with specialized statistical methods, graphics and data-management tools. Packages are distributed through repositories such as CRAN.

## 6.2 A Short History

R derives from the S language developed by John Chambers and colleagues at Bell Laboratories. S began in 1976 as an internal statistical analysis environment built around Fortran libraries; in 1988 it was rewritten in C and began to resemble its later form. The book *Statistical Models in S* by Chambers and Hastie (the “white book”) is identified as a key description of its statistical modeling functionality.

R itself was created by Ross Ihaka and Robert Gentleman at the University of Auckland. Its development milestones include 1993 as the first public announcement, 1995 as the adoption of the GNU General Public License following Martin Mächler’s contribution, 1996 as the creation of public R-help and R-developer mailing lists, 1997 as the formation of the R Core Group, and 2000 as the public release of R 1.0.0.

### Remark: Package masking

When packages are attached, names such as `add`, `convolve`, `spectrum`, `frame` and `save.image` can be masked by objects from other attached packages. This illustrates a practical feature of the R search path: attaching packages can introduce functions with identical names, and the most recently attached binding can mask another.

## 6.3 General Features of R

R’s design closely reflects statistical work. Objects, functions, packages and applications can be created within the environment. R is platform-independent, free under GNU licensing, and can integrate with



systems such as Hadoop/HDFS and with languages including C, C++, Python, Java, FORTRAN and JavaScript. The user community supports educational resources and *The R Journal*.

## 6.4 R versus RStudio

### Key Concept: Language versus IDE

R is the programming language/runtime that performs computations. RStudio is an integrated development environment that provides a more convenient interface around the language.

The interface includes a source editor, console, environment pane, file browser, data viewer and plotting panel.

## 6.5 Installation and Ways to Execute Code

Install R from <https://cloud.r-project.org> and install RStudio separately as an optional IDE. Commands can be executed in three ways:

1. type directly in the R console and press Return;
2. write an .R script, select one or more lines, and execute them with Ctrl+Return or the Run command;
3. execute a code chunk from an R Markdown document using its Run control.

Listing 6.1: A numerical vector, its mean and its sample standard deviation.

```
x <- c(8, 10, 5, 3, 11)
mean(x)
sd(x)
```

The mean is 7.4; the standard deviation uses the sample divisor  $n - 1$ . The clustering chapter develops a multivariate covariance example.

## 6.6 Learning Resources

- <https://cran.r-project.org/doc/manuals/r-release/R-intro.pdf>
- <https://education.rstudio.com/learn/>
- RStudio's getting-started material on R Markdown
- <https://journal.r-project.org>

### Takeaway: Quick review

R supplies statistical computation and a programming language; RStudio supplies a development interface. Scripts make analyses repeatable, and packages extend the available methods. Understanding the formulas remains necessary for interpreting the output.

# Distances and Cluster Analysis

---

Distance measures express how observations differ across several variables. Their scale and geometry determine the groups discovered by cluster analysis. This chapter develops proximity measures, data transformations, K-means and criteria for selecting the number of clusters.

## 7.1 Covariance in R

The following R example computes variances, pairwise covariances and a full sample covariance matrix:

Listing 7.1: Individual variances and covariances agree with the corresponding entries of the sample covariance matrix.

```
set.seed(2026)
vect1 = rnorm(10, 1, 2)
vect2 = rnorm(10, 1.5, 1.5)
vect3 = rnorm(10, 2, 1)
dataset = cbind(vect1, vect2, vect3)
dataset = as.matrix(dataset)
var(vect1); var(vect2); var(vect3)
cov(vect1, vect2); cov(vect1, vect3); cov(vect2, vect3)
cov(dataset)
```

The output depends on the random draw. The seed makes this simulation reproducible. The functions `var` and `cov` use the sample divisor  $n - 1$ , rather than the population divisor  $N$ .

## 7.2 Similarity, Dissimilarity and Proximity

A **similarity** is larger when objects are more alike and often lies in  $[0, 1]$ . A **dissimilarity** is smaller when objects are more alike; its minimum is often zero while its upper bound depends on the measure. **Proximity** covers either similarity or dissimilarity.

### 7.2.1 Similarity-matrix example

In a classical example from Kruskal and Wish (1978, p. 31), students rated the cultural similarity of 12 countries from 1 (very different) to 9 (very similar). The lower-triangular matrix contains the mean ratings; larger values indicate greater perceived similarity.

Table 7.1: Mean cultural-similarity ratings on a scale from 1 (very different) to 9 (very similar). Countries: 1 Brazil, 2 Congo, 3 Cuba, 4 Egypt, 5 France, 6 India, 7 Israel, 8 Japan, 9 China, 10 Russia, 11 U.S.A., 12 Serbia.

|    | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  | 11  | 12 |
|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|----|
| 1  |     |     |     |     |     |     |     |     |     |     |     |    |
| 2  | 4.8 |     |     |     |     |     |     |     |     |     |     |    |
| 3  | 5.3 | 4.6 |     |     |     |     |     |     |     |     |     |    |
| 4  | 3.4 | 5.0 | 5.2 |     |     |     |     |     |     |     |     |    |
| 5  | 4.7 | 4.0 | 4.1 | 4.8 |     |     |     |     |     |     |     |    |
| 6  | 4.5 | 4.8 | 4.0 | 5.8 | 3.4 |     |     |     |     |     |     |    |
| 7  | 3.8 | 3.3 | 3.6 | 4.7 | 4.0 | 4.1 |     |     |     |     |     |    |
| 8  | 3.5 | 3.4 | 2.9 | 3.8 | 4.2 | 4.5 | 4.8 |     |     |     |     |    |
| 9  | 2.4 | 4.0 | 5.5 | 4.4 | 3.7 | 4.1 | 3.0 | 4.2 |     |     |     |    |
| 10 | 3.1 | 3.4 | 5.4 | 4.4 | 5.1 | 4.5 | 4.2 | 4.6 | 5.7 |     |     |    |
| 11 | 5.4 | 2.4 | 3.2 | 3.3 | 5.9 | 4.3 | 5.9 | 6.1 | 2.6 | 5.0 |     |    |
| 12 | 3.2 | 3.5 | 5.1 | 4.3 | 4.7 | 4.0 | 4.4 | 4.3 | 5.1 | 6.7 | 3.6 |    |

## 7.3 Distance Measures

### 7.3.1 Minkowski distance

For vectors  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  and order  $m \geq 1$ , the Minkowski distance is

$$d_m(\mathbf{x}, \mathbf{y}) = \left( \sum_{i=1}^n |x_i - y_i|^m \right)^{1/m}. \quad (7.1)$$

$\mathbf{x}, \mathbf{y}$  vectors being compared;

$x_i, y_i$   $i$ th components;

$n$  vector dimensionality;

$m$  Minkowski order;

$d_m$  resulting dissimilarity.

The Euclidean norm is

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2}, \quad (7.2)$$

and Euclidean distance is

$$d_2(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}. \quad (7.3)$$

Setting  $m = 1$  gives Manhattan or city-block distance,

$$d_1(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n |x_i - y_i|. \quad (7.4)$$

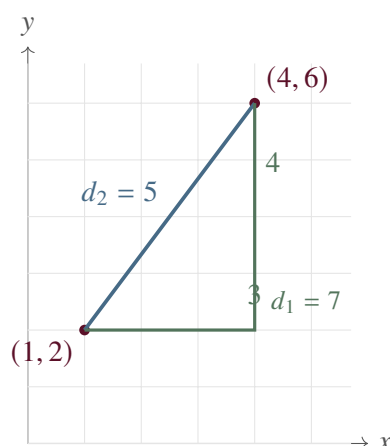


Figure 7.1: Euclidean and Manhattan distances between  $(1, 2)$  and  $(4, 6)$ : a straight-line distance of 5 versus an axis-aligned distance of 7.

### 7.3.2 Mahalanobis distance

For column vectors  $\mathbf{x}$  and  $\mathbf{y}$  and a positive-definite covariance matrix  $\Sigma$ , Mahalanobis distance is

$$d_M(\mathbf{x}, \mathbf{y}) = \sqrt{(\mathbf{x} - \mathbf{y})^\top \Sigma^{-1} (\mathbf{x} - \mathbf{y})}. \quad (7.5)$$

$\Sigma$  covariance matrix describing scale and correlation among components;

$\Sigma^{-1}$  inverse covariance matrix;

$d_M$  distance after accounting for statistical variation.

When  $\Sigma = I$ , Mahalanobis distance reduces to Euclidean distance. Covariance scaling reduces the contribution of directions with large natural variability and accounts for correlations between coordinates. A singular covariance matrix has no ordinary inverse; redundant variables require a modified treatment before this formula can be used.

### 7.3.3 Why the covariance matrix matters

The inverse covariance matrix changes the geometry of the space before the straight-line distance is evaluated. Its role can be summarized in four connected effects:

- **Scale adjustment.** Variables measured in different units are rescaled so that a numerically large unit does not dominate merely because of its scale.
- **Correlation adjustment.** Directions containing redundant, strongly correlated information are downweighted rather than counted repeatedly.
- **Whitening interpretation.** Conceptually, the data are transformed toward a coordinate system with unit variance and zero linear correlation; Euclidean distance is then measured in that transformed geometry.
- **Elliptical geometry.** Directions in which the data naturally vary a great deal are treated as less surprising than directions with little natural variation. Equal-distance contours are therefore ellipsoids in the original coordinates rather than circles or spheres.

**Key Concept: Euclidean versus Mahalanobis geometry**

Euclidean distance treats coordinate directions as equally scaled and mutually orthogonal. Mahalanobis distance instead measures separation relative to the covariance structure of the data. A displacement along a high-variance direction contributes less than an equally large displacement along a low-variance direction.

## 7.4 What Clustering Tries to Achieve

Clustering assigns objects to groups so that objects within a cluster are as similar as possible while objects in different clusters are as dissimilar as possible according to a chosen proximity measure. Applications include marketing, time-series analysis, insurance, machine learning, data mining, pattern recognition, image analysis, genomics and systems biology. It is an unsupervised method: the aim is to discover structure rather than learn a predefined input–output target.

Clustering supports identifying group structure, profiling units such as customers, and finding patterns that are not immediately recognizable.

## 7.5 Data Transformations Before Clustering

Distance-based methods are sensitive to scale. Possible transformations include:

- center each variable by subtracting its mean;
- standardize using `scale()`, centering each column and dividing by its sample standard deviation to obtain mean 0 and standard deviation 1;
- log transform strictly positive values when multiplicative differences are relevant;
- replace values by ranks;
- apply no transformation when the original scale is appropriate.

For a centered vector of  $n > 1$  non-missing values, the scale divisor is

$$x_{\text{rms}} = \sqrt{\frac{1}{n-1} \sum_{i=1}^n x_i^2}. \quad (7.6)$$

The correct choice depends on the dataset and on which differences should count as substantively important. With default centering, the displayed divisor equals the sample standard deviation. Without centering, `scale(x, center = FALSE)` applies the same  $n - 1$  root-sum-of-squares convention to uncentered values, rather than the usual root mean square with divisor  $n$ . Constant columns cannot be standardized to unit variance.

## 7.6 K-means Clustering

Choose a number of clusters  $K$  and partition the  $n$  observations into disjoint sets  $C_1, \dots, C_K$  whose union contains all observations. K-means seeks a partition with low within-cluster variation.

For a nonempty cluster  $C_k$ , its within-cluster sum of squares can be expressed through pairwise distances as

$$W(C_k) = \frac{1}{2|C_k|} \sum_{i, i' \in C_k} \|X_i - X_{i'}\|_2^2. \quad (7.7)$$

This equals

$$\sum_{i \in C_k} \|X_i - \bar{X}_k\|_2^2, \quad (7.8)$$

Here each  $X_i$  is a  $p$ -dimensional observation vector,  $|C_k|$  is the cluster size, and  $\bar{X}_k = |C_k|^{-1} \sum_{i \in C_k} X_i$  is the centroid. Clusters must be nonempty.

The double sum in [equation \(7.7\)](#) counts each unordered pair twice. The factor  $1/(2|C_k|)$  makes it exactly equal to the centroid sum of squares in [equation \(7.8\)](#).

The global objective is

$$W(K) = \sum_{k=1}^K \sum_{i \in C_k} \|X_i - \bar{X}_k\|_2^2. \quad (7.9)$$

The objective minimizes within-cluster dispersion under squared Euclidean distance. Implementations must handle empty clusters and resolve nearest-centroid ties consistently.

Because exhaustive search over all partitions is computationally impractical, a standard iterative algorithm is used:

1. choose  $K$  and initialize a cluster assignment;
2. compute the centroid of every current cluster;
3. assign each observation to the nearest centroid in Euclidean distance;
4. repeat centroid computation and reassignment until the assignments stop changing.

Each iteration decreases or leaves unchanged the K-means objective, but different initializations can converge to different local solutions. Run the algorithm from multiple starting points and retain the solution with the smallest within-cluster sum of squares.

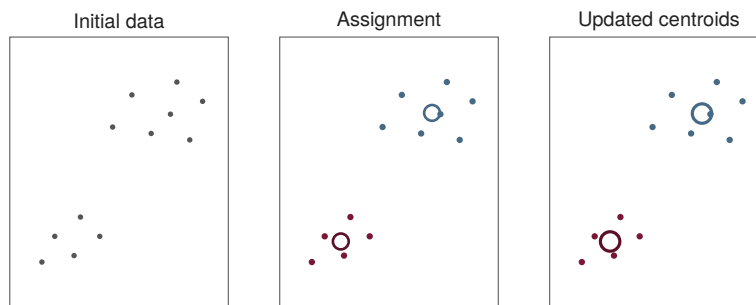


Figure 7.2: Schematic K-means assignment and centroid update. Filled points are observations; open circles are centroids. Coordinates are illustrative.

## 7.7 Choosing the Number of Clusters

Choosing  $K$  affects interpretation: concluding that a market contains two customer profiles instead of three can change managerial interpretation.

### 7.7.1 Elbow method

Because optimal  $W(K)$  cannot increase as  $K$  increases, the elbow method looks for a point after which additional clusters yield relatively small reductions in within-cluster variation.

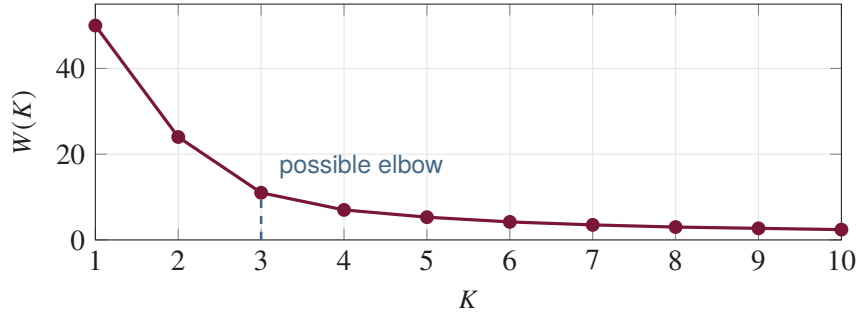


Figure 7.3: Schematic elbow criterion: diminishing reductions in within-cluster variation suggest a candidate value of  $K$ . Values are illustrative.

### 7.7.2 Between-cluster variation

Between-cluster variation is defined by

$$B(K) = \sum_{k=1}^K |C_k| \|\bar{X}_k - \bar{X}\|_2^2, \quad (7.10)$$

where  $\bar{X}_k$  is cluster  $k$ 's centroid and  $\bar{X}$  is the grand mean,

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i. \quad (7.11)$$

The total sum of squares decomposes as  $T = W(K) + B(K)$ , where  $T = \sum_i \|X_i - \bar{X}\|_2^2$ . For optimal partitions, increasing  $K$  cannot increase  $W(K)$ , so  $B(K)$  cannot decrease. Maximizing  $B(K)$  alone therefore does not select a useful number of clusters.

### 7.7.3 Calinski–Harabasz index

The Calinski–Harabasz index combines between-cluster separation and within-cluster compactness:

$$CH(K) = \frac{B(K)/(K-1)}{W(K)/(n-K)}. \quad (7.12)$$

The selected number of clusters is

$$\hat{K} = \arg \max_{K \in \{2, \dots, K_{\max}\}} CH(K). \quad (7.13)$$

The index of Calinski & Harabasz (1974) is defined for  $2 \leq K < n$  and  $W(K) > 0$ ; choose  $K_{\max} < n$ . Larger values indicate stronger separation relative to within-cluster spread, after adjusting for the numbers of clusters and observations. The criterion is particularly informative for compact clusters of comparable sizes. A maximum at  $K = 2$ , for example, selects a two-cluster partition among the candidates considered.

### 7.7.4 Gap statistic

The Gap statistic of Tibshirani et al. (2001) compares observed dispersion with dispersion under a reference distribution without cluster structure. A uniform distribution over a suitable data-derived region is a common reference. For  $M$  simulated reference datasets, define

$$\text{Gap}(K) = \frac{1}{M} \sum_{b=1}^M \log W_b^*(K) - \log W(K), \quad (7.14)$$

Here  $W(K)$  is the observed within-cluster sum of squares and  $W_b^*(K)$  is the corresponding value for reference replicate  $b$ , fitted using the same clustering procedure. The average is taken after applying the logarithm. If  $s_{\log W^*}(K)$  is the sample standard deviation of the  $M > 1$  reference log-dispersions, the simulation-adjusted uncertainty is  $s(K) = s_{\log W^*}(K) \sqrt{1 + 1/M}$ . All dispersions must be positive. The one-standard-error rule selects

$$\hat{K} = \min \{K : \text{Gap}(K) \geq \text{Gap}(K + 1) - s(K + 1)\}. \quad (7.15)$$

A large positive Gap value means that the observed partition is more compact than partitions of the reference data. The one-standard-error rule favors the smallest number of clusters whose improvement is statistically comparable to that at the next value of  $K$ .

#### Takeaway: Quick review

Scaling defines cluster geometry; K-means minimizes within-cluster dispersion. Because the objective can converge to different local solutions, multiple initializations are useful. Choosing  $K$  requires separation or reference criteria, since optimal within-cluster dispersion cannot increase as  $K$  grows.



# Hierarchical Clustering and Linkage Methods

---

Hierarchical clustering builds a nested sequence of partitions rather than requiring one fixed number of clusters at the start. The central design choice is the *linkage*: the rule used to translate point-to-point dissimilarities into a dissimilarity between whole clusters. The resulting hierarchy is visualized with a dendrogram, whose horizontal cuts define candidate clusterings.

## 8.1 From K-means to Hierarchical Clustering

K-means fits exactly  $K$  groups and can depend on the random initialization of centroids or assignments. Agglomerative hierarchical clustering takes a different route: it starts with every observation in its own cluster and repeatedly merges clusters until all observations belong to one group. The complete hierarchy can therefore be computed before deciding where to cut it.

### Key Concept: Agglomerative clustering

Given point-to-point dissimilarities  $d_{ij}$  and a linkage function  $d(G, H)$  between clusters  $G$  and  $H$ :

1. start with  $n$  singleton clusters;
2. find the pair of current clusters with the smallest linkage dissimilarity;
3. merge that pair;
4. repeat until only one cluster remains.

The sequence of merges is deterministic once the dissimilarities, linkage, and tie-breaking convention are fixed.

A **divisive** hierarchy proceeds in the opposite direction, starting from one cluster and recursively splitting it. The development here focuses on the agglomerative case.

### 8.1.1 A seven-point merge sequence

For seven observations, one possible agglomerative sequence is

$$\begin{aligned}
 &\{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}, \{7\} \\
 &\longrightarrow \{1, 2\}, \{3\}, \{4\}, \{5\}, \{6\}, \{7\} \\
 &\longrightarrow \{1, 2\}, \{3\}, \{5\}, \{4, 7\}, \{6\} \\
 &\longrightarrow \{1, 2, 6\}, \{3\}, \{5\}, \{4, 7\} \\
 &\longrightarrow \{1, 2, 4, 6, 7\}, \{3\}, \{5\} \\
 &\longrightarrow \{1, 2, 3, 4, 6, 7\}, \{5\} \\
 &\longrightarrow \{1, 2, 3, 4, 5, 6, 7\}.
 \end{aligned}$$

Different linkage definitions can generate different merge sequences even when the underlying observations are identical.

## 8.2 Dendrograms and Cutting the Hierarchy

A **dendrogram** is a tree representation of the agglomerative sequence. Leaves correspond to individual observations, internal nodes correspond to merges, and the vertical coordinate records the dissimilarity or merge cost associated with each fusion. Cutting the tree horizontally produces a partition: every connected component below the cut becomes one cluster.

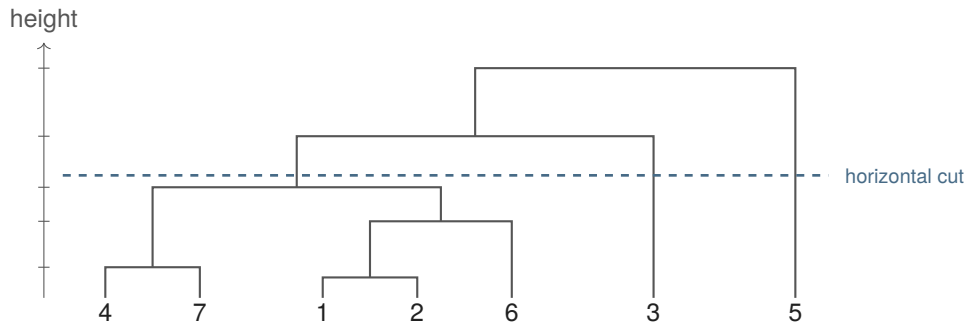


Figure 8.1: Schematic dendrogram for the seven-point merge sequence. A horizontal cut determines the number and membership of the resulting clusters. Heights are illustrative.

A large vertical gap between consecutive merge heights is often interpreted as evidence for a natural boundary: a cut inside that gap separates groups that can be merged internally at relatively low cost but require a much larger cost to combine with one another.

### 8.2.1 Ward's variance-increase criterion

In Ward-style hierarchical clustering, clusters are merged so that the increase in total within-cluster sum of squares is as small as possible. If  $C_i$  and  $C_j$  have centroids  $\bar{X}_i$  and  $\bar{X}_j$ , then the merge cost can be written as

$$\Delta W(C_i, C_j) = \frac{|C_i| |C_j|}{|C_i| + |C_j|} \|\bar{X}_i - \bar{X}_j\|_2^2. \quad (8.1)$$

Equivalently,

$$\Delta W = W(C_i \cup C_j) - W(C_i) - W(C_j). \quad (8.2)$$

|                                 |                                                                |
|---------------------------------|----------------------------------------------------------------|
| $ C_i ,  C_j $                  | numbers of observations in the two clusters;                   |
| $\bar{X}_i, \bar{X}_j$          | cluster centroids;                                             |
| $\ \bar{X}_i - \bar{X}_j\ _2^2$ | squared Euclidean separation between centroids;                |
| $\Delta W$                      | increase in within-cluster sum of squares caused by the merge. |

Under this convention, low fusion heights indicate clusters that can be combined with little loss of compactness, whereas high fusion heights indicate more distinct groups. Ward's method therefore tends to favor compact, approximately spherical clusters.

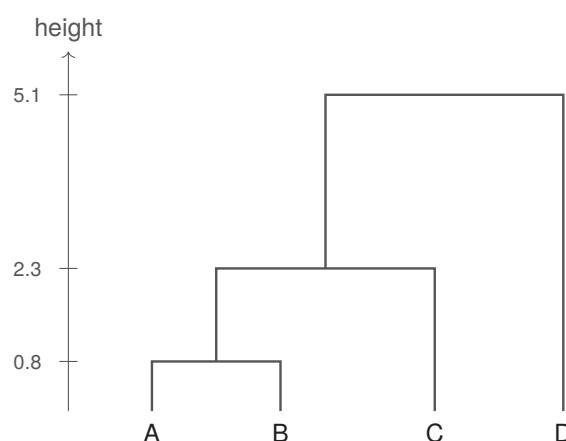


Figure 8.2: Reading dendrogram heights: small merge heights indicate similar groups; a large jump suggests that the groups joined above that level are substantially more dissimilar under the chosen criterion.

## 8.3 Linkage Functions

Let  $d_{ij}$  denote the dissimilarity between observations  $X_i$  and  $X_j$ . A **linkage** is a function  $d(G, H)$  that converts the collection of cross-cluster dissimilarities into one dissimilarity score for clusters  $G$  and  $H$ . The linkage determines which two clusters are merged next.

### 8.3.1 Single linkage

Single, or nearest-neighbor, linkage uses the closest cross-cluster pair:

$$d_{\text{single}}(G, H) = \min_{i \in G, j \in H} d_{ij}. \quad (8.3)$$

At a dendrogram cut height  $h$ , single linkage can be interpreted through connectivity: observations belong to the same cluster when they can be connected by a chain of pairwise links whose dissimilarities do not exceed  $h$ . This makes single linkage capable of discovering elongated shapes, but it also creates the classic **chaining** problem: one short bridge can join two otherwise diffuse groups.

### 8.3.2 Complete linkage

Complete, or farthest-neighbor, linkage uses the most dissimilar cross-cluster pair:

$$d_{\text{complete}}(G, H) = \max_{i \in G, j \in H} d_{ij}. \quad (8.4)$$

If a complete-linkage dendrogram is cut at height  $h$ , every pair of observations inside a resulting cluster has dissimilarity at most  $h$ . This strongly favors compact clusters. Its characteristic limitation is **crowding**: a point can be closer to observations in another cluster than to some observations in its own cluster, so clusters may be compact without being well separated.

### 8.3.3 Average linkage

Average linkage uses all cross-cluster pairs:

$$d_{\text{average}}(G, H) = \frac{1}{|G||H|} \sum_{i \in G} \sum_{j \in H} d_{ij}. \quad (8.5)$$

It often provides a compromise between the local sensitivity of single linkage and the diameter-oriented behavior of complete linkage. However, a cut of an average-linkage tree has no equally simple pairwise interpretation, and the resulting hierarchy can change under a monotone increasing transformation of the original dissimilarities.

### 8.3.4 Centroid linkage

When observations are vectors in  $\mathbb{R}^p$  and Euclidean geometry is appropriate, centroid linkage compares group means:

$$d_{\text{centroid}}(G, H) = \|\bar{X}_G - \bar{X}_H\|_2. \quad (8.6)$$

Centroid linkage is intuitive and closely related to the centroid idea in K-means, but it can generate **inversions**: a later merge can occur at a lower dendrogram height than an earlier merge. The resulting tree therefore need not have monotonically increasing fusion heights.

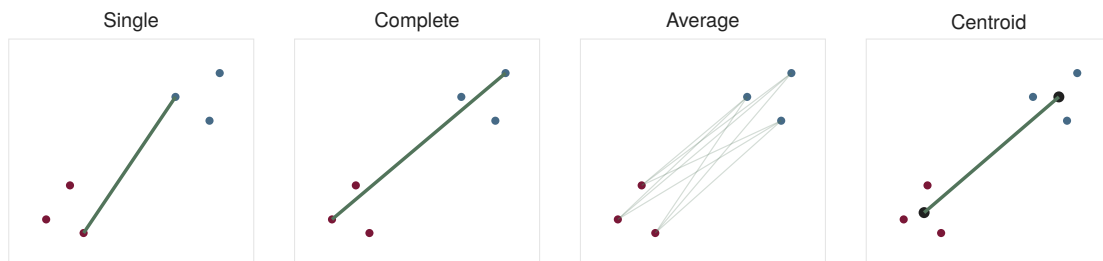


Figure 8.3: Geometric intuition for four linkages. Single uses the closest cross-cluster pair, complete the farthest pair, average all cross-cluster pairwise dissimilarities, and centroid the distance between cluster means.

## 8.4 Comparing the Main Linkages

The linkages differ not only in their formula but also in the structure they tend to produce.

Single, complete, and average linkage operate directly on the pairwise dissimilarities  $d_{ij}$  and, under their standard agglomerative definitions, produce dendrograms without inversions. Single and complete linkage are unchanged by strictly increasing transformations of the dissimilarities because they depend only on the ordering of minima or maxima. Average linkage is not generally invariant because averaging transformed distances can change which cluster pair has the smallest score.

There is no universally best linkage. The choice should reflect the geometry that is substantively meaningful: connectivity favors single linkage, tight diameter control favors complete linkage, average

Table 8.1: Qualitative comparison of the principal linkage rules.

| Linkage         | No inversions | Invariant to monotone transform | Simple cut interpretation | Characteristic issue                                                       |
|-----------------|---------------|---------------------------------|---------------------------|----------------------------------------------------------------------------|
| <b>Single</b>   | Yes           | Yes                             | Yes                       | Chaining; clusters can become long and diffuse.                            |
| <b>Complete</b> | Yes           | Yes                             | Yes                       | Crowding; compact groups need not be well separated.                       |
| <b>Average</b>  | Yes           | No                              | No                        | Balanced behavior, but weaker cut interpretation and transform invariance. |
| <b>Centroid</b> | No            | No                              | No                        | Inversions can make later fusion heights decrease.                         |

linkage offers a compromise, Ward favors compact variance-based groups, and centroid linkage offers a mean-based interpretation at the cost of possible inversions.

#### Takeaway: Quick review

Hierarchical agglomerative clustering repeatedly merges the two closest clusters according to a chosen linkage and records the sequence in a dendrogram. A horizontal cut turns the hierarchy into a partition. Ward linkage minimizes the increase in within-cluster sum of squares; single, complete, average and centroid linkage define cluster dissimilarity in different ways and therefore can produce materially different groupings.

# Formula and Concept Quick Reference

This reference collects the principal formulas and their essential conditions. Use the chapter discussions for notation, interpretation and worked examples; population and sample conventions remain distinct.

Table 9.1: Principal relationships for descriptive statistics, association, regression and clustering.

| Concept                     | Core relationship                                                                      |
|-----------------------------|----------------------------------------------------------------------------------------|
| Relative frequency          | $f_i = n_i/N$ , with $\sum_i f_i = 1$ .                                                |
| Percentage frequency        | $p_i = 100f_i$ , with $\sum_i p_i = 100$ .                                             |
| Cumulative frequency        | $N_i = \sum_{j=1}^i n_j$ .                                                             |
| Arithmetic mean             | $\mu = N^{-1} \sum_i x_i$ .                                                            |
| Weighted mean               | $\mu_w = (\sum_i x_i w_i) / (\sum_i w_i)$ .                                            |
| Median                      | Middle ordered position; for even quantitative $N$ , average the two central values.   |
| Variance                    | $\sigma^2 = N^{-1} \sum_i (x_i - \mu)^2$ .                                             |
| Standard deviation          | $\sigma = \sqrt{\sigma^2}$ .                                                           |
| Coefficient of variation    | $CV = \sigma/\mu$ , for ratio-scale data with $\mu > 0$ .                              |
| z-score                     | $z_i = (x_i - \mu)/\sigma$ , with $\sigma > 0$ .                                       |
| Gini impurity               | $E_1 = 1 - \sum_i f_i^2$ , normalized by $k/(k-1)$ .                                   |
| Expected independence count | $n_{ij}^* = n_{i \cdot} n_{\cdot j} / N$ .                                             |
| Pearson $\chi^2$            | $\sum_{ij} (n_{ij} - n_{ij}^*)^2 / n_{ij}^*$ .                                         |
| Normalized $\chi^2$         | $\chi^2 / [N \min(h-1, k-1)]$ , the square of Cramér's $V$ .                           |
| Covariance                  | $\sigma_{xy} = N^{-1} \sum_i (x_i - \mu_x)(y_i - \mu_y)$ .                             |
| Pearson correlation         | $\rho_{xy} = \sigma_{xy} / (\sigma_x \sigma_y)$ .                                      |
| OLS slope                   | $b = \sigma_{xy} / \sigma_x^2 = \rho \sigma_y / \sigma_x$ .                            |
| OLS intercept               | $a = \bar{y} - b\bar{x}$ .                                                             |
| Matrix product / transpose  | $(AB)_{ij} = \sum_\ell a_{i\ell} b_{\ell j}$ ; $(AB)^\top = B^\top A^\top$ .           |
| Determinant / inverse       | For $2 \times 2$ matrices, $\det A = ad - bc$ ; $AA^{-1} = I$ requires full rank.      |
| Eigenpair / SVD             | $Ax = \lambda x$ , $x \neq 0$ ; $A = U\Sigma V^\top$ with nonnegative singular values. |
| Minkowski distance          | $d_m = (\sum_i  x_i - y_i ^m)^{1/m}$ , $m \geq 1$ .                                    |
| Euclidean distance          | $d_2 = \sqrt{\sum_i (x_i - y_i)^2}$ .                                                  |
| Manhattan distance          | $d_1 = \sum_i  x_i - y_i $ .                                                           |
| Mahalanobis distance        | $[(x - y)^\top \Sigma^{-1} (x - y)]^{1/2}$ , for positive-definite $\Sigma$ .          |
| K-means objective           | $W(K) = \sum_k \sum_{i \in C_k} \ X_i - \bar{X}_k\ ^2$ .                               |
| Between-cluster variation   | $B(K) = \sum_k  C_k  \ \bar{X}_k - \bar{X}\ ^2$ .                                      |
| Calinski–Harabasz           | $CH = [B/(K-1)] / [W/(n-K)]$ , $2 \leq K < n$ , $W > 0$ .                              |
| Gap statistic               | $M^{-1} \sum_b \log W_b^*(K) - \log W(K)$ , with the one-standard-error rule.          |

| Concept          | Core relationship                                                            |
|------------------|------------------------------------------------------------------------------|
| Ward merge cost  | $\Delta W = \frac{ C_i  C_j }{ C_i  +  C_j } \ \bar{X}_i - \bar{X}_j\ _2^2.$ |
| Single linkage   | $d(G, H) = \min_{i \in G, j \in H} d_{ij}.$                                  |
| Complete linkage | $d(G, H) = \max_{i \in G, j \in H} d_{ij}.$                                  |
| Average linkage  | $d(G, H) = ( G  H )^{-1} \sum_{i \in G} \sum_{j \in H} d_{ij}.$              |
| Centroid linkage | $d(G, H) = \ \bar{X}_G - \bar{X}_H\ _2.$                                     |

Clustering Decision Guide

|                                          |                                                                                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Goal or data geometry                    | Practical starting point                                                                                                                      |
| Fixed number of compact groups           | K-means, with standardized variables when measurement scales are not directly comparable; use several initializations.                        |
| Nested structure or exploratory grouping | Agglomerative hierarchical clustering; select a linkage that reflects the intended cluster geometry, then inspect meaningful dendrogram cuts. |
| Connectivity or elongated shapes         | Single linkage, while watching for chaining.                                                                                                  |
| Tight cluster diameter                   | Complete linkage, while watching for crowding.                                                                                                |
| Compromise between single and complete   | Average linkage.                                                                                                                              |
| Compact variance-based groups            | Ward’s criterion, based on the increase in within-cluster sum of squares.                                                                     |
| Mean-based cluster comparison            | Centroid linkage, with awareness that dendrogram inversions can occur.                                                                        |

Table 9.2: A compact decision guide for the clustering methods developed in the preceding chapters.

# Mathematical Toolkit

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This toolkit collects elementary algebra, functions, set operations and calculus used in statistical work. Each topic supplies a mathematical building block for interpreting formulas and understanding multivariate methods.

## A.1 Factorials and Polynomial Identities

The factorial of a positive integer is

$$n! = n(n-1)(n-2) \cdots 2 \cdot 1. \quad (\text{A.1})$$

Anagrams illustrate factorials: the three distinct letters of “POT” have  $3! = 6$  permutations. For repeated letters with multiplicities  $n_1, \dots, n_k$  summing to  $n$ , the number of distinct anagrams is  $n!/(n_1! \cdots n_k!)$ . By convention,  $0! = 1$ .

Useful binomial identities include

$$\begin{aligned} (a+b)^2 &= a^2 + 2ab + b^2, \\ (a-b)^2 &= a^2 - 2ab + b^2, \\ (a+b)^3 &= a^3 + 3a^2b + 3ab^2 + b^3, \\ (a-b)^3 &= a^3 - 3a^2b + 3ab^2 - b^3. \end{aligned}$$

For three terms,

$$\begin{aligned} (a+b+c)^2 &= a^2 + b^2 + c^2 + 2ab + 2ac + 2bc, \\ (a+b-c)^2 &= a^2 + b^2 + c^2 + 2ab - 2ac - 2bc, \\ (a-b+c)^2 &= a^2 + b^2 + c^2 - 2ab + 2ac - 2bc, \\ (a-b-c)^2 &= a^2 + b^2 + c^2 - 2ab - 2ac + 2bc. \end{aligned}$$

Pascal’s (Tartaglia’s) triangle generates coefficients for powers of a binomial. For example,

$$(a+b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5. \quad (\text{A.2})$$

The coefficients 1, 5, 10, 10, 5, 1 are the fifth row of binomial coefficients, counting the top row as row zero.



## A.2 Functions, Domain and Codomain

A function transforms an input into an output. For example,

$$y = f(x) = |x|. \quad (\text{A.3})$$

Its domain is  $\mathbb{R}$  and its range is  $[0, +\infty)$ . The codomain must be specified: choosing  $[0, +\infty)$  makes this mapping surjective, whereas choosing  $\mathbb{R}$  does not. The absolute value maps both  $-5$  and  $5$  to  $5$ .

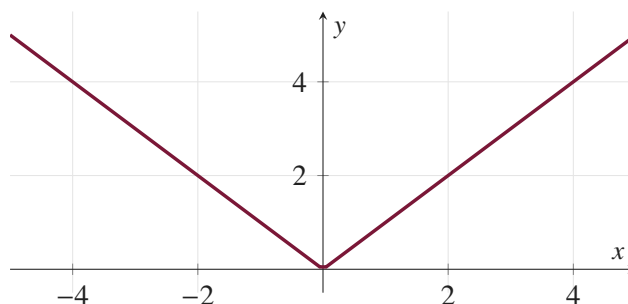


Figure A.1: Graph of  $y = |x|$ , the function used in the mathematical recap.

For  $a < b$ , interval notation includes open  $(a, b)$ , closed  $[a, b]$ , half-open  $[a, b)$  and  $(a, b]$  intervals, as well as unbounded intervals.

### A.2.1 Injective, surjective and bijective mappings

Mappings are classified by which inputs and outputs they connect:

- an **injective** function never sends two distinct inputs to the same output;
- a **surjective** function reaches every element of the codomain;
- a **bijective** function is both injective and surjective and therefore has a two-sided inverse;
- a relation that assigns one input to more than one output is not a function.

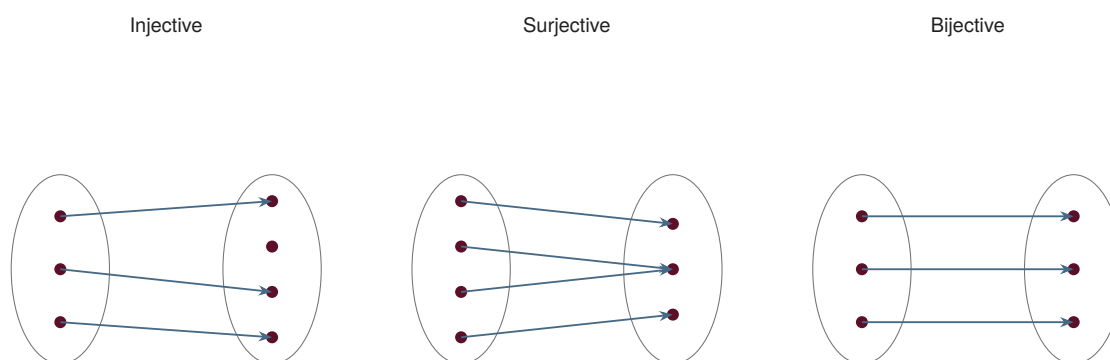


Figure A.2: Injective, surjective and bijective mappings: uniqueness of preimages and coverage of the codomain.

### A.3 Inverse, Exponential and Logarithmic Functions

For a one-to-one function  $y = f(x)$ , the inverse satisfies  $f^{-1}(y) = x$ . For example, on the restricted domain  $x \geq 0$ , the inverse of  $y = x^2$  is  $x = \sqrt{y}$ .

The natural exponential and logarithm are inverse functions:

$$f(x) = e^x, \quad f^{-1}(x) = \ln x,$$

where  $e \approx 2.718281828$ . Real logarithms require positive arguments. The exponential maps  $\mathbb{R}$  onto  $(0, +\infty)$ , and the logarithm is its inverse on that range. Useful identities include

$$\begin{aligned} e^0 &= 1, & e^{-x} &= \frac{1}{e^x}, & (e^x)^y &= e^{xy}, \\ e^{x+y} &= e^x e^y, & e^{x-y} &= \frac{e^x}{e^y}, & \ln(x^y) &= y \ln x, \\ \ln 1 &= 0, & \ln(1/x) &= -\ln x, & \ln(xy) &= \ln x + \ln y, \\ \ln(xy) &= \ln x + \ln y, & \ln(x/y) &= \ln x - \ln y. \end{aligned}$$

### A.4 Set Theory

Let  $\Omega$  denote a reference set. Common set operations are:

- **union**  $A \cup B$ : elements belonging to  $A$  or  $B$  (or both);
- **intersection**  $A \cap B$ : elements belonging to both;
- **disjoint sets**:  $A \cap B = \emptyset$ ;
- **complement**  $\bar{A}$ : elements of  $\Omega$  not in  $A$ ;
- **subset**  $B \subset A$ : every element of  $B$  belongs to  $A$ ;
- **difference**  $A - B$ : elements of  $A$  not belonging to  $B$ .

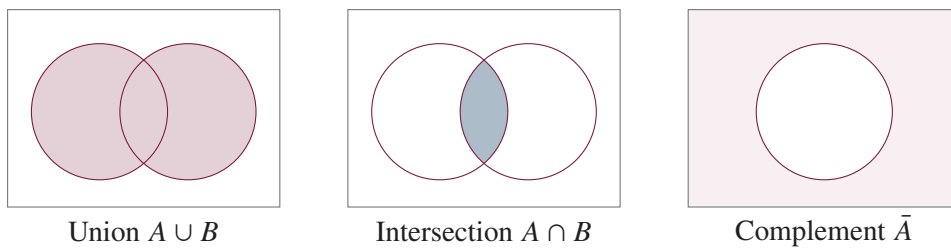


Figure A.3: Union, intersection and complement: shaded regions identify the resulting sets.

### A.5 Limits and Derivatives

The derivative of  $f$  is defined by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}, \quad (\text{A.4})$$

whenever the limit exists. Geometrically,  $f'(x_0)$  is the slope of the tangent line at  $x_0$ .

The  $\varepsilon$ - $\delta$  definition states that  $f(x) \rightarrow L$  as  $x \rightarrow x_0$  when, for every  $\varepsilon > 0$ , there exists  $\delta > 0$  such that  $0 < |x - x_0| < \delta$  implies  $|f(x) - L| < \varepsilon$ .

For  $f(x) = x^2$  at  $x_0 = 1$ ,

$$f'(1) = \lim_{h \rightarrow 0} \frac{(1+h)^2 - 1}{h} = \lim_{h \rightarrow 0} (2+h) = 2.$$

The derivative value  $f'(1) = 2$  is a tangent slope. The function limit is a different quantity:  $\lim_{x \rightarrow 1} x^2 = 1$ .

## A.6 Definite Integrals

The definite integral is the limit of sums of signed rectangular areas:

$$\int_a^b f(x) dx = \lim_{m \rightarrow \infty} \sum_{j=1}^m f(\xi_j) \Delta x, \quad \Delta x = \frac{b-a}{m}. \quad (\text{A.5})$$

For  $a < b$ , divide  $[a, b]$  into  $m$  equal subintervals and select  $\xi_j$  in subinterval  $j$ . For continuous  $f$ , the limit is independent of those choices. The integral represents signed area and equals geometric area when  $f \geq 0$ .

For continuous  $f$ , find an antiderivative  $F$  with  $F' = f$ ; then

$$\int_a^b f(x) dx = F(b) - F(a). \quad (\text{A.6})$$

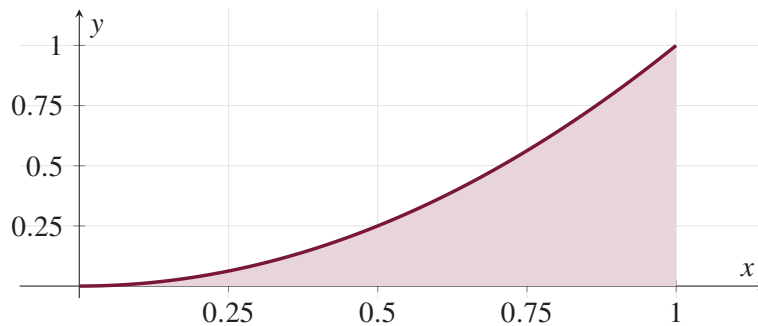


Figure A.4: The shaded area under  $y = x^2$  on  $[0, 1]$  equals  $\int_0^1 x^2 dx = 1/3$ .

A probability-density function is nonnegative and has total integral one:  $\int_{-\infty}^{+\infty} f(x) dx = 1$ .

# References and Further Reading

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The following books, research references and computing resources support the topics developed in these notes.

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